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Faculty of Forestry and Wood technology

Department of Forest Ecology



DOCTORAL THESIS

**Long-term effect of changes in underground water level for
vitality and productivity of the floodplain forests**

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Study Program: Forest Ecology

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DISSERTATION THESIS TOPIC

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Title of the thesis: **Long-term effect of changes in underground water level for vitality and productivity of the floodplain forests**

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Guides to writing a thesis:

1. The objective of the research is to investigate how long-term shifts in groundwater regimes—driven by climate change and anthropogenic hydrological modifications—impact the vitality and growth dynamics of floodplain forest ecosystems. The study focuses on forests where the disconnection of rivers from their floodplain systems has led to measurable declines in groundwater levels.
2. The research combines ecological field monitoring, dendrochronological analysis, and wood anatomical assessments to evaluate how dominant tree species such as Pedunculate oak and Narrow-leaved ash respond to declining subsurface water availability and increased drought frequency.
3. Special emphasis will be placed on comparing species-specific physiological adaptations and growth responses under current and projected hydrological conditions. Furthermore, the study aims to assess the vulnerability and resilience of typical floodplain species from local to broader geographic scales, identifying spatial patterns of sensitivity to hydrological stress.
4. The work provides critical insights into the current state of floodplain forests and contributes valuable guidance for ecological research and forest management. It supports the development of strategies for hydrological restoration, adaptive species selection, and long-term resilience planning to sustain riparian woodland ecosystems under changing environmental conditions.

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Aknowledgements

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***“The roots below the earth claim no rewards for making
the branches fruitful”***

~ Rabindranath Tagore

List of publications

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- I. **Basu, S***, Stojanović, M., Jevšenak, J., Buras, A., Kulhavý, J., Hornová, H., Světlík, J., 2024. Pedunculate oak is more resistant to drought and extreme events than narrow-leaved ash in Central European floodplain forests. *For. Ecol. Manag.* 561, 121907. <https://doi.org/10.1016/j.foreco.2024.121907>
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into management strategies to sustain productivity and resilience in future floodplain forests under changing climates.

lze vytvořit strategii hospodaření, která umožní udržet produktivitu a zajistit přežití lužních lesů v podmínkách měnícího se klimatu.

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1. Introduction

1.1 Climate Change Impacts on Forest Ecosystems

Climate change is one of the greatest challenges for the environment this century. It is primarily driven by anthropogenic activities leading to the temperatures rise and its weather patterns shift. Since the end of the 1800s, the mean global temperature has risen by approximately 1.1 °C (IPCC, 2023). Europe, described as a temperature hotspot, has experienced an even greater increase of about 2.0 °C (IPCC, 2023; Lewis et al., 2019). In recent decades, annual temperatures have been rising by roughly 0.2 °C per (IPCC, 2023).

At the same time, the patterns of rainfall are changing. For example, summer rainfall is declining, and increased variability marked by intense, short-duration events and prolonged dry spells (Spinoni et al., 2018, 2015). Summer rainfall has declined by up to 20% in parts of Europe, while rainfall is becoming more shorter, erratic and more intense events are separated by longer dry periods (IPCC, 2008). These shifts reduce water availability during key growth periods, particularly in spring and early summer hindering plant water uptake and limiting carbon assimilation through photosynthesis (Mundim and Pringle, 2018).

Among all terrestrial ecosystems, forests are considered one of the most vulnerable to climate change, due to their long-life cycles, complex structure, and close dependence on climatic and hydrological stability. These disturbances affect forests both directly and indirectly. Though higher temperatures promotes growth (Treml et al., 2025); it may also raise evapotranspiration, leading to water stress and reduced growth. Increased Vapor pressure deficit (VPD) makes stomata close earlier, which slows down photosynthesis and lowers production (Trotsiuk et al., 2021). Climate change alters important phenological events like flowering and leaf-out (Ford et al., 2016; Xu et al., 2023). Changes in the water cycle are becoming more obvious at the same time. The changes reduce groundwater recharge and river flows, placing added pressure on forest ecosystems (Rottler et al., 2020; van Tiel et al., 2024). In Central and Eastern Europe, water scarcity and droughts are becoming more common (Cook et al., 2020; Skiadaresis et al., 2019). Cumulatively, these stresses weaken forests, making them more susceptible to large-scale disturbances (Seidl et al., 2017). Warmer, drier years favour wildfires, drought, and insect outbreaks, while

warmer, wetter years lead to wind damage and disease. Between 1979 and 2018, roughly 33.4 billion tonnes of forest biomass in Europe were affected by disturbances; wind causing the highest losses (40%), followed by fire (34%) and pests (26%) (Forzieri et al., 2021). The cumulative effect of these factors is causing shifts in forest composition and changes in forest structure. Drought tolerant species are diverging in their distribution range while the sensitive species are slowly declining. Because forest ecosystems respond over long timescales, their capacity to adjust to the rapidly accelerating pace of climate change is limited. This temporal mismatch can compromise their physiological function, reduce carbon assimilation, hinder regeneration, and lead to biodiversity loss.

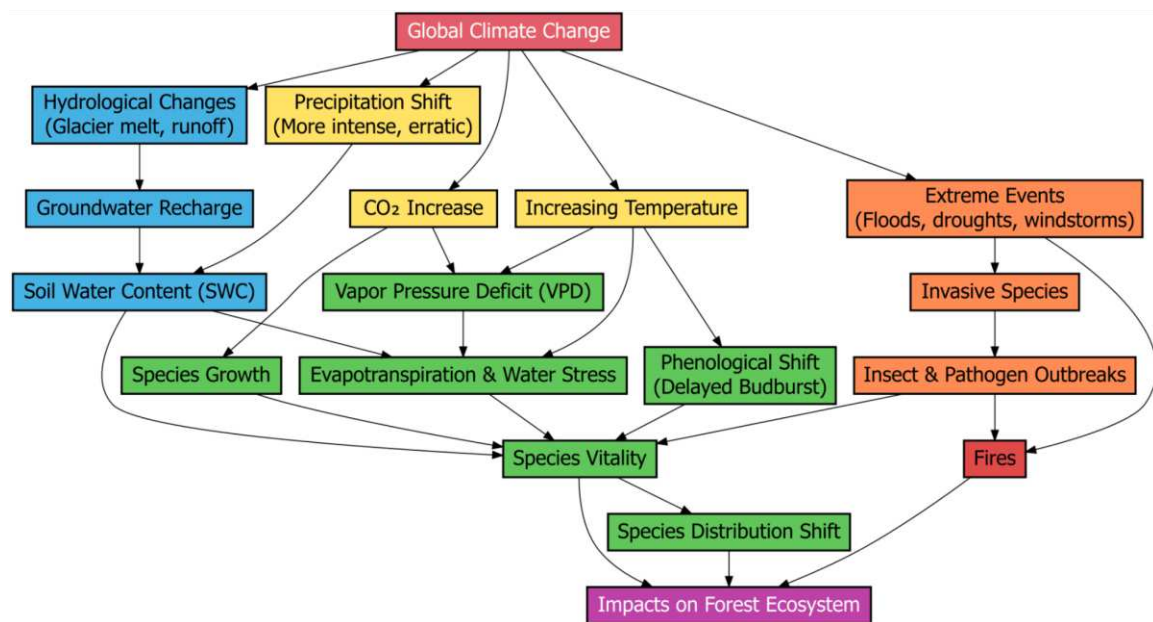


Figure 1. Potential impacts of global climate change on forest ecosystems.

1.2 Floodplain-forest ecosystems and their hydrological dependence

1.2.1 Floodplain forests

Floodplain forests, sometimes referred to as lowland or riparian forests, are among the most dynamic and hydrologically dependent ecosystems. They occur along riverbanks that are periodically flooded, where variable water levels influence their composition and ecological roles (Hughes and Rood, 2003; Machar et al., 2020). Their ecological functions are strongly linked to seasonal flooding, sediment deposition, and groundwater flow. These processes influence nutrient cycling, soil aeration, and species dispersion, hence promoting a diverse range of flora and fauna (Hughes and Rood, 2003; Klimo and Hager, 2001).

A characteristic feature of floodplain forests is their fertile soils, enriched with alluvium from periodic floods. The sediments have various textures, from coarse sandy loams near riverbanks to finer silty clays in low-lying areas of the floodplain, creating varied microhabitats (Klimo and Prax, 1985). The periodic wetting and drying limit organic matter formation, thereby maintaining a balance of aeration and moisture.

European floodplain forests are typically categorised into two primary types, softwood and hardwood, according to their location within the floodplain and their susceptibility to floods (Klimo and Hager, 2008; Klimo et al., 2013). Usually, softwood floodplain woods are found near active river channels on lower elevation locations and undergo recurrent and extended flooding, typically persisting for several weeks annually (Mikulová et al., 2020). Consequently, they are predominantly occupied by rapidly proliferating pioneer species, including willows (*Salix* spp.) and poplars (*Populus* spp.), which have a high tolerance for wet environments (Klimo and Hager, 2001; Mikulová et al., 2020). These trees display certain morphological adaptations, such as adventitious roots and developed aerenchyma tissue, facilitating oxygen delivery and root viability in anaerobic soils (Mozo et al., 2021). Their stable growth and structural flexibility enable them to withstand fluctuating water levels and to regenerate quickly after hydrological disturbances.

In contrast, hardwood floodplain forests are found on higher ground within the floodplain, where flooding is less frequent and of shorter duration. These forests are characterised by slower-growing, long-lived species such as oak, ash, and elm (Klimo et al., 2013). These

trees prefer moist yet well-drained soils and exhibit heightened sensitivity to extended water saturation, although they depend on elevated groundwater levels to sustain development during arid intervals (Garamszegi et al., 2022). Although hardwood species exhibit less tolerance to severe flooding compared to softwoods, they play a crucial role in enhancing the structural complexity and ecological stability of floodplain forests (Basu et al., 2024; Szatniewska et al., 2022). Their extensive root systems and prolonged lifespans strengthen drought resilience and facilitate persistent carbon sequestration throughout time.

The distribution of these forest types frequently indicates delicate changes in microtopography, soil composition, and flood duration. In a single floodplain system, there is often a gradient of species composition, transitioning from softwood communities adjacent to the riverbank to hardwood forests located further inland (Penka et al., 1991). This zonation accommodates a diverse range of plant and animal species; each adapted to certain hydrological circumstances. The coexistence of both forest types augments biodiversity, resilience, and ecological function throughout the floodplain landscape (Penka et al., 1991). Acting as natural buffers, floodplain forests moderate hydrological extremes by trapping sediments, and filtering pollutants. In addition, they serve as ecological corridors, supporting species migration and maintaining connectivity across fragmented landscapes.

Historically, floodplain forests were abundant along Europe's major rivers, including the Danube, Rhine, Elbe, and Morava, playing a vital role in regional biodiversity and ecological processes. Floodplain forests provide essential biological functions, providing habitats for numerous species and enabling crucial ecosystem services such as carbon sequestration, flood management, and water filtration (Klimo and Hager, 2008; Machar, 2008).

1.2.2 Groundwater - floodplain ecosystem interaction

Floodplain ecosystems are intricately linked to underlying groundwater systems. These connections are shaped by sediment structure, hydrological variability, and ecological processes (Klimo and Hager, 2001). The interaction between surface water and groundwater governs key functions such as soil moisture retention and nutrient transport (Irvine et al., 2024). The magnitude and direction of surface-groundwater exchange are influenced by sediment composition, which varies from fine-grained clays to coarse gravels (McGarr et al.,

2024). The layers affect permeability and water circulation, resulting in spatially heterogeneous recharge and discharge areas within the floodplain (Larocque et al., 2016).

In several floodplains, groundwater has been increased by river flooding and localised precipitation (Lemon and Keim, 2025). During the early growing season, shallow groundwater is mostly affected by precipitation; however, later in summer, river water levels have a more pronounced influence (Lemon and Keim, 2025). This dynamic exchange is multidirectional, fluctuating throughout daily, seasonal, and annual cycles, with flow paths influenced by terrain, vegetation, and stratified sediment layers.

Groundwater plays a vital role in supporting vegetation, particularly under drought stress (Skiadaresis et al., 2019). These attributes facilitate the stability of productivity amid hydrological extremes. However, if groundwater levels decline faster than root systems can adjust, the viability and survival of trees are jeopardized (Imada et al., 2008).

Land use change, urbanisation, deforestation, and channel modification have altered recharge dynamics across Europe (Riedel and Weber, 2020). These alterations diminish infiltration and decrease groundwater levels, which, over time, may restrict tree water supply and heighten ecological stress (Klimo and Prax, 1985). Severe events, such as extended droughts, can cause significant groundwater reductions, hastening discharge and diminishing long-term storage even in hydrologically abundant areas.

The complexity of groundwater flow is further enhanced by internal floodplain features such as macropores, aquifer compartmentalisation, and stratified sediments (Lemon and Keim, 2025; Ó Dochartaigh et al., 2019). These effects lead to uneven water distribution, creating a mosaic of moisture availability across the floodplain. This diversity promotes hydrological niche separation among plant species, allowing coexistence via depth preferences and water-use strategies (Sarkar et al., 2019).

Comprehending these interactions is fundamental for the sustainable management of floodplain forests. The resilience of these ecosystems depends not only on surface hydrology, but also on stable and accessible groundwater systems. As climate change alters rainfall patterns and increases evapotranspiration, preserving the integrity of surface–groundwater linkages become increasingly urgent.

1.2.3 Global and regional threats to groundwater and floodplain forests

Groundwater and floodplain forests are more susceptible globally and regionally due to anthropogenic activities and climate change. Excessive groundwater extraction for agriculture, urban supplies, and industry has led to declining water tables, increased drought stress, and forest dieback in several lowland ecosystems. The impacts are now evident in several locations, including Europe and Australia (Alao et al., 2024; Stigter et al., 2023).

Climate change is intensifying pressures on groundwater-dependent forest ecosystems by increasing drought frequency, raising temperatures, and altering precipitation patterns. These changes reduce aquifer recharge and increase water loss through evapotranspiration, ultimately decreasing groundwater availability (Pool et al., 2021). As a result, these ecosystems face growing instability, which undermines their essential functions such as climate regulation, biodiversity maintenance, and ecosystem productivity (Anderegg et al., 2020). Research indicates that ecosystems reliant on groundwater are particularly vulnerable to these changes (Hughes et al., 2012), and globally around 50 percent of floodplain forests are already degraded, with even higher levels of degradation reported in Europe and the United States.

Historically, wet forests such as swamp, bog, alluvial, spring, and floodplain forests were widespread across Central Europe, sustained by consistent groundwater levels. Pollen analyses reveal that hardwood floodplain forests were among the most dominant habitats in the mid-Holocene, prior to significant human intervention (Leuschner and Ellenberg, 2017). However, these forest types have become increasingly rare due to both anthropogenic pressures and environmental change (Hughes et al., 2012). Nowadays, approximately 50 percent of Europe's population lives on floodplains (Tockner et al., 2008), yet these landscapes have seen severe decline

In regions such as Central and Southern Europe, the situation is particularly critical. During prolonged dry periods, demand for groundwater rises sharply. When this is combined with lower recharge and continued pumping, it erodes the natural buffering role of groundwater (Skiadarensis et al., 2019). This dual demand weakens floodplain forests and presents wider threats to water security and regional resilience. Lall et al., (2020) indicate that groundwater depletion is inadequately addressed, despite its significant socioeconomic and ecological consequences. The management and modification of rivers, construction of dams, and loss

of lateral connectivity have impeded overbank flooding, which was previously a crucial process for replenishing groundwater (Klimo and Hager, 2001). The Danube along with other significant European rivers, have experienced diminished floodplain interchange and decreased riverbeds, disrupting connections between rivers and aquifers (Funk et al., 2019).

These hydrological alterations result in distinct ecological consequences. Trees subjected to extended water stress exhibit physiological deterioration, characterised by diminished shoot water potential, early leaf drop, and increased mortality (Stovall et al., 2019). Over time, changes in species composition, a decline in biodiversity, and increased vulnerability to pests and diseases become apparent (Berthelot et al., 2015). Such degradation also affects the global carbon cycle. If groundwater stress continues to be overlooked, floodplain forests may transition from carbon sinks to net carbon sources (Mitchard, 2018). Moreover, groundwater quality is progressively affected by contamination, introducing additional complication to conservation efforts (Wang et al., 2024).

1.2.4 South Moravian Floodplain Forest: Condition and Risks

The floodplain forest in South Moravia is a significant surviving element of riparian ecosystems in Central Europe (Klimo et al., 2013). Located near the confluence of the Morava and Dyje Rivers in the southeast of the Czech Republic, this forest is part of a transitional climate zone that is impacted by both continental and atlantic climatic patterns (Machar, 2008; Unar and Šamonil, 2008). Warm summers and cold winters, along with consistent precipitation throughout the year, contribute to the unique ecological character of the area. Numerous plant species, such as *Fraxinus angustifolia*, *Fraxinus excelsior*, *Quercus robur*, *Ulmus laevis*, and *Populus alba*, are supported by rich cambic and gleyic fluvisols (Penka et al., 1991). Additionally, there is an herbaceous understory that includes *Leucojum aestivum* and *Lysimachia vulgaris*. The ecosystem offers essential services including carbon sequestration, flood mitigation, biodiversity enhancement, and water filtration (Klimo et al., 2013).

Human activities have significantly influenced this landscape. Archaeological evidence indicates that increased land use started in the early Middle Ages, especially after Slavic population in the sixth century AD (Klimo and Hager, 2001). By the time of the Great Moravian Empire, forest clearance and grazing had altered the hydrological balance,

increasing flood peaks. During medieval times, further waves of colonisation, deforestation, and habitat fragmentation changed the forest structure (Klimo and Hager, 2001). Rapid-growing poplars and softwoods often replaced native hardwoods, particularly in response to human pressures and climatic fluctuations (Klimo and Hager, 2008).

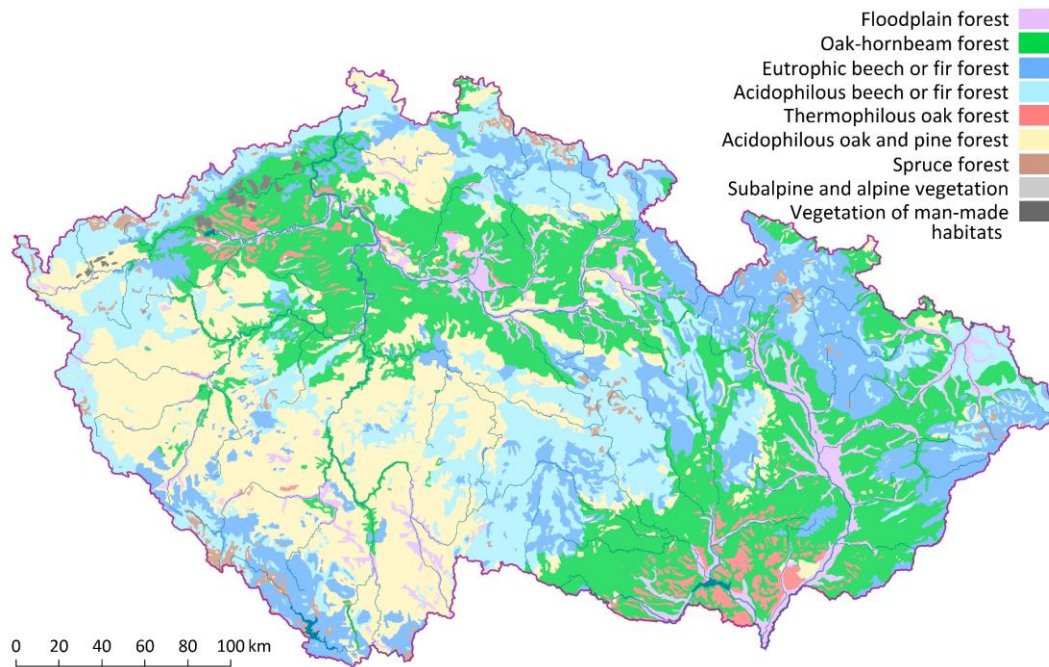


Figure 2. Floodplain forest distribution in the Czech Republic (Neuhäuslová-Novotná, 1998).

The 19th and 20th centuries brought industrialisation and major river engineering. Between 1948 and the late 20th century, more than 67 km, about 20% of the Morava River was straightened, and over 25% of its active branches were severed. This reduced flooding frequency and disrupted the floodplain's natural dynamics. Industrial pollutants, combined with untreated urban waste, further degraded the ecosystem (Klimo et al., 2013). Hydrology was desynchronised by the construction of large reservoirs in the Nové Mlýny region, which attenuated downstream floods.

At the same time, industrial pollution and untreated urban effluent were discharged into the river system, progressively degrading water quality and the surrounding ecosystems. By the 1950s, the Morava was regarded as one of Europe's most contaminated rivers, with more than 70% of the adjacent territory classified as biologically compromised (Klimo and Hager, 2008).

From the 1960s onwards, agricultural intensification accelerated ecological change. Hedgerows were systematically removed, meadows drained, and pastures converted into large monocultures, leading to soil compaction, habitat loss, and a continuing decline in groundwater levels (Penka et al., 1991). The impacts of these transformations were not uniform. Certain areas experienced partial ecological recovery following the post-war departure of German agriculturalists, which allowed abandoned land to revert to forest and meadow (Penka et al., 1991). By contrast, urbanised zones saw forests supplanted by agricultural land, further contributing to water quality deterioration. The pressures imposed by historical land use have been compounded by climatic extremes. The severe drought of 2018 sharply reduced water availability, pushing trees to their physiological limits, while groundwater tables have dropped by more than a metre in some areas (Kowalska et al., 2020).

The ecological balance of the South Moravian floodplain forests has been profoundly altered, beginning with the degradation of soil properties. Decades of river regulation, drainage, and channelization have disrupted the natural flooding regime, causing formerly hydric, nutrient-rich alluvial soils to become compacted, less aerated, and depleted of organic matter (Sedlák et al., 2023). The absence of regular flood pulses has reduced sediment deposition, leading to nutrient shortages in many areas, while nitrogen has accumulated unevenly in patches of disturbed land (Marchetti et al., 2020). These soil changes have undermined water storage capacity, nutrient cycling, and the ecological stability of the entire forest system.

Weakened soil has left trees more susceptible to disease outbreaks and pests. Ash dieback caused by *Hymenoscyphus fraxineus* is devastating ash populations (Jankovský and Holdenrieder, 2009), while oak dieback and root pathogens are increasingly afflicting oak (Doležal et al., 2010), the dominant canopy species. These conditions weaken mature trees and create opportunities for secondary pests, further accelerating forest decline.

This decline feeds into a severe regeneration problem. Natural recruitment of oak has fallen sharply because the altered hydrology no longer clears competing vegetation or provides suitable germination niches. In response, drought- and shade-tolerant species such as hornbeam and field maple are spreading, reshaping species composition and simplifying

forest structure (Janik et al., 2011, 2008). As a result, habitat diversity is eroding, and the forests' ability to provide vital ecosystem services, including carbon sequestration, water retention, and nutrient cycling is diminishing.

On top of these challenges, additional pressures are altering the landscape. Invasive species like *Robinia pseudoacacia* have spread into disturbed areas, changing soil chemistry and outcompeting native vegetation (Vítková et al., 2017) , while extensive *Populus* plantations, especially short-rotation coppices, have replaced parts of the natural forest (Hobza, 2007). These plantations consume large amounts of water and add to soil drying, further undermining the resilience of the remaining native stands.

Channel reconnection, and pool re-establishment have all been part of restoration efforts in areas such as Soutok since the 1990's (Klimo and Hager, 2001) . These have facilitated the preservation of uncommon species, enhanced habitat connectivity, and contributed to the restoration of biological processes. In addition, artificial flooding has been regularly implemented in regions such as Pohansko since 2011 (Bártová et al., 2021), although its effects on forest growth remain poorly studied. Currently, protected places like Litovelské Pomoraví and Pálava preserve the remaining forest pieces (Klimo and Hager, 2008).

1.3 Responses of Floodplain Trees to Water Deficit: Insights from oak and ash

How floodplain tree species respond to fluctuating and increasingly limited water availability is central to understanding their long-term survival and the resilience of these ecosystems. Short-term water deficit in ash causes immediate stomatal closure, rapid leaf wilting, and accelerated leaf senescence due to its shallow plate root system and endomycorrhizal dependency, which restrict water uptake to surface soil layers (Koprowski et al., 2018; Szatniewska et al., 2022). When drought persists, ash exhibits marked reductions in stem radial increment and frequently fails to sustain new root elongation, resulting in poor access to deep moisture and low resilience to groundwater decline. In contrast, oak employs a deep taproot system, extensive fine roots, and ectomycorrhizal symbiosis to draw water from both deep groundwater and, after intense rain events, from moist surface horizons

(Allen, 2015). This dual foraging strategy supports sustained transpiration and photosynthesis during moderate drought, but when multi-year groundwater deficit occurs, even oaks show declining growth and reduced ring width (Stojanović et al., 2015).

Hydraulic architecture and stomatal regulation together determine the drought resilience of these species. Oak develops wide earlywood vessels that enable high rates of water transport and vigorous transpiration under wet conditions, but these vessels are highly susceptible to drought-induced embolism. When cavitation blocks water flow in winter or spring, oak must restore hydraulic conductivity by producing new, large-diameter vessels at the onset of the next growing season (Koprowski et al., 2018; Roibu et al., 2020). This hydraulic risk is offset by an anisohydric stomatal strategy: oak maintains gas exchange and allows leaf water potential to decline during moderate drought, sustaining photosynthetic activity but increasing the risk of xylem failure if water deficit intensifies. In contrast, ash develops narrower vessels that resist cavitation but limit water transport. For instance, vessel diameters in *Fraxinus excelsior* typically remain below 50 μm , reflecting an anatomical strategy common to species adapted to drier environments or with shallow rooting systems (Anfodillo et al., 2013). This narrow diameter not only constrains axial water flow but may also enhance embolism resistance, as narrower conduits tend to have fewer and smaller pit areas, reducing vulnerability to drought-induced failure (Anfodillo and Olson, 2021). Its isohydric stomatal behavior, characterized by rapid closure to protect xylem integrity, leads to a quick decrease in stomatal conductance and carbon gain as soon as soil moisture begins to decline (Szatniewska et al., 2022)

These morphological and physiological differences result in distinct growth and phenological responses to water scarcity (Nguyen, 2021). Both species have shown decreased leaf area, early leaf fall, delayed bud burst, and decreased reproductive output when subjected to prolonged drought (Nosenko et al., 2025; Polgar and Primack, 2011; Vitasse et al., 2019). Nevertheless, oak exhibits significant adaptability by reallocating resources towards root development and diminishing shoot growth during deficits, so partially mitigating losses in stem growth (Abrams, 1990). Ash, on the other hand, shows abrupt, low-plasticity reductions in growth and canopy development, with only limited or delayed recovery even after the return of favorable moisture conditions (Kowalska et al., 2020; Szatniewska et al., 2022).

Recurrent water deficit affects both species by lowering the development of new vessels and depleting starch stores, but it also makes ash particularly vulnerable to several infections, which causes dieback even in cases of mild drought (Janik et al., 2011). For both species, a prolonged water deficit directly results in hydraulic failure, radial growth stagnation, and loss of regeneration capacity, according to experimental and field studies conducted throughout Central and Eastern Europe. Ash exhibits more immediate mortality, while oak shows a gradual, trait-mediated decline (Heklau et al., 2019; Roibu et al., 2020; Šenfeldr et al., 2021). Because oak occurs across a wide climatic gradient in Europe, from humid northern to drought-prone southern regions, populations may differ in their drought resilience and growth sensitivity depending on local climatic conditions.

1.4 Tree-ring and dendroanatomy approaches in forest research

1.4.1 Tree-ring analysis

The annual growth increments that result from secondary growth in trees, known as tree rings, have long been used as a fundamental stand-in for studying forest dynamics and ecological processes (Fritts, 1976). Dendroecology, which utilises tree-ring series to address ecological enquiries, has significantly enhanced our comprehension of tree and forest responses to both episodic disturbances and progressive environmental changes (Amoroso et al., 2017; Fritts and Swetnam, 1989). The majority of early dendroecological research was conducted in the Northern Hemisphere's temperate zones, but in recent decades, this method has been extended to boreal, Mediterranean, and tropical forests, demonstrating its adaptability to a variety of biomes worldwide (Amoroso et al., 2017; Rozendaal and Zuidema, 2011). A lifetime of physiological reactions to climate, competition, and disturbance is captured by each tree ring core, which encodes environmental information from the year of its creation. This makes tree rings remarkably valuable due to their length and annual resolution (Cook and Kairiukstis, 1990; Speer, 2010).

Tree-ring widths features are now commonly acknowledged as trustworthy proxy of tree vitality and productivity, reflecting the cumulative effects of biotic and climatic stressors

as well as the processes of carbon allocation and assimilation (Babst et al., 2014; Dobbertin, 2005). Historically, dendroecology examined climate-growth interactions by linking tree-ring chronologies with monthly or seasonal climate variables, providing significant insights into the influence of large-scale climatic variability on tree growth (Fritts, 1976; Jacoby and D'Arrigo, 1997). The incorporation of high-resolution, daily climate data into dendroecological analysis enables the identification of short-term events such as heatwaves, late frosts, and brief droughts can substantially impact radial growth but would be hidden in coarser temporal datasets (Babst et al., 2014). This transformation has facilitated a more detailed characterisation of species-specific climate sensitivity. Moreover, it has revealed rapid and complex growth responses to climatic extremes, thereby advancing our understanding of the physiological mechanisms underlying tree productivity and resilience.

Crucially, tree-ring data have become instrumental in operationalising ecological resilience frameworks, such as the one proposed by Lloret et al., (2011), which defines resilience through three primary metrics: resistance (the degree of growth reduction during disturbance), recovery (the speed and magnitude of post-disturbance growth return), and resilience (overall capacity to regain pre-disturbance growth levels). These parameters can be directly quantified from tree-ring chronologies, offering a retrospective lens into how trees and forests have historically responded to stressors. By applying this framework, researchers can assess the capacity of trees to withstand and bounce back from disturbances such as droughts, insect outbreaks, or extreme climatic events. However, as highlighted by Schwarz et al. (2020), this approach can oversimplify complex growth–disturbance relationships and is sensitive to methodological choices such as the selection of reference periods, which can strongly influence calculated indices. They further emphasise that the method may fail to capture delayed or non-linear growth responses, potentially overlooking important aspects of long-term resilience dynamics.

High-resolution methodologies, when integrated with conventional cross-dating and measurement of increment cores, have enhanced the capacity of dendroecology to evaluate sudden disturbances, prolonged growth patterns, and the resilience or susceptibility of trees and forests in the context of rapid climate change (Cailleret et al., 2019; Gazol et al., 2015).

1.4.2. Dendroanatomy

In contrast to traditional tree-ring analysis, which predominantly indicates the cumulative result of seasonal growth, wood anatomy enables researchers to investigate the detailed properties of xylem development and their reactions to certain environmental stimuli. Both moderate, long-term environmental trends and abrupt shifts in climatic conditions, such as severe drought, frost, or heatwaves, can influence these anatomical traits. Crucially, without a thorough analysis of their species-specific hydraulic strategies; how they behave and shape their vessels in response to environmental stresses, the climatic vulnerability of some tree species may go unnoticed. An important step forward in dendroecological study is the growing ability to look at how wood's structural features change from year to year over longer periods of time (Fonti et al., 2010; Gärtner et al., 2015; von Arx et al., 2016). Anatomical properties preserved within tree rings, such as vessel diameter, density, and wall thickness, are shaped by a complex interplay of factors, including genetic background, hormonal regulation, and, most importantly, environmental conditions during wood formation (Chave et al., 2009; Fonti et al., 2010; Rathgeber et al., 2016; Tulik, 2014). Climate directly influences xylem cell features, while extreme events such as droughts can indirectly alter wood structure by impacting the tree's physiological condition and the mobilisation of photo assimilates (Fonti et al., 2010; Fonti and Jansen, 2012).

Due to the importance of water in plant function and survival, tree hydraulics, as shown by xylem anatomical diversity, is increasingly crucial for understanding how trees respond to environmental change. Interspecific variations in hydraulic properties are well understood, but intraspecific variation is still relatively little studied, even though it is important for forecasting how various species and populations may adapt to the effects of climate change (Anderegg and Meinzer, 2015; Fonti et al., 2013). Quantifying the number, size, and placement of xylem channels tells us a lot about how well water moves through the tree stem. The Hagen-Poiseuille equation states that hydraulic conductivity increases exponentially with conduit diameter, hence slight changes in lumen size can significantly increase water transport capacity (Tyree and Ewers, 1991). This efficiency, however, incurs a cost, as large vessels are particularly susceptible to embolism—especially during

drought or frost events—which can significantly compromise xylem functionality (Hacke, 2014; Hacke et al., 2006; Sperry et al., 2008).

Trees regulate leaf gas exchange by moving water from soil to canopy, which links xylem function to photosynthesis and plant productivity (Landsberg et al., 2017). Hydraulic architecture may promote and constrain plant function by creating physical growth boundaries (Brodribb and Cochard, 2009; Sperry et al., 2008). The complex relationship between xylem anatomy, physiological processes, and the environment highlights the importance of wood anatomical studies in addressing pressing questions about tree and population vulnerability to climate change, particularly considering increasing drought frequency and severity (Anderegg and Meinzer, 2015). While interspecific differences in hydraulic strategies are well documented, the extent of intraspecific variability in xylem anatomical traits under fluctuating hydrological conditions remains poorly understood, particularly for co-occurring species such as oak and ash. Understanding this variability is essential for assessing their potential to adjust plastically to changing water regimes.

2. Aim and hypothesis

The aim of this study was to evaluate how oak and ash respond to fluctuations in groundwater availability and drought across local and continental scales, and to identify the mechanisms underlying their resilience. As discussed in the introduction, floodplain forests are strongly dependent on groundwater, yet the extent to which variations in water tables influence species-specific growth sensitivity remains insufficiently quantified. In this context we hypothesised (H1) that fluctuations in groundwater levels significantly affect the growth sensitivity of oak and ash to climate-driven water supply, including precipitation, groundwater levels, and drought indices such SPEI.

Because of seasonality of river runoff, we further hypothesised (H2) that the coupling between tree growth and water availability differs across the growing season.

Given the contrasting physiological strategies of the species, with oak characterised by deep rooting and anisohydric behaviour and ash by shallow rooting and isohydric stomatal regulation, we hypothesised (H3) that oak demonstrates greater drought resistance, recovery, and resilience than ash.

Although both oak and ash are ring-porous species, their contrasting water-use strategies are expected to drive divergent anatomical adjustments under fluctuating water regimes. We therefore hypothesised (H4) that ash exhibits greater variability in vessel traits than oak, reflecting its more conservative and environmentally sensitive hydraulic strategy.

As water availability influences vessel formation, but the effect of flooding on xylem development in mature trees is not well understood, we hypothesised (H5) that artificial flooding increases earlywood vessel size and hydraulic conductivity in ash, but not in oak, reflecting species-specific water using strategies.

Finally, because oak spans a wide climatic gradient from humid northern to drought-prone southern Europe, we hypothesised (H6) that populations from southern regions exhibit higher resilience and distinct climate sensitivity compared to northern populations.

3. Material and methodology

3.1 Study site

3.1.1 Czech Republic:

Our study carried out in specified plots situated within the floodplain forest at the convergence of the Morava and Dyje rivers (Fig. 3a and Fig.3b).

The Lednice site was established as a permanent investigation plot and has not been subject to any forest management since 1970. Pohansko has seen limited management interventions in the past twenty years, notably the implementation of artificial spring flooding in 2011. Soutok is a regularly maintained forest, following a forest management plan, whereas the Lanzhot site is an ICOS (Integrated Carbon Observation System) study site that has remained unmanaged for the past 40 years. The long-term (1961–2010) mean annual precipitation across all locations was 506 mm, and the mean annual temperature was 9.5 °C.

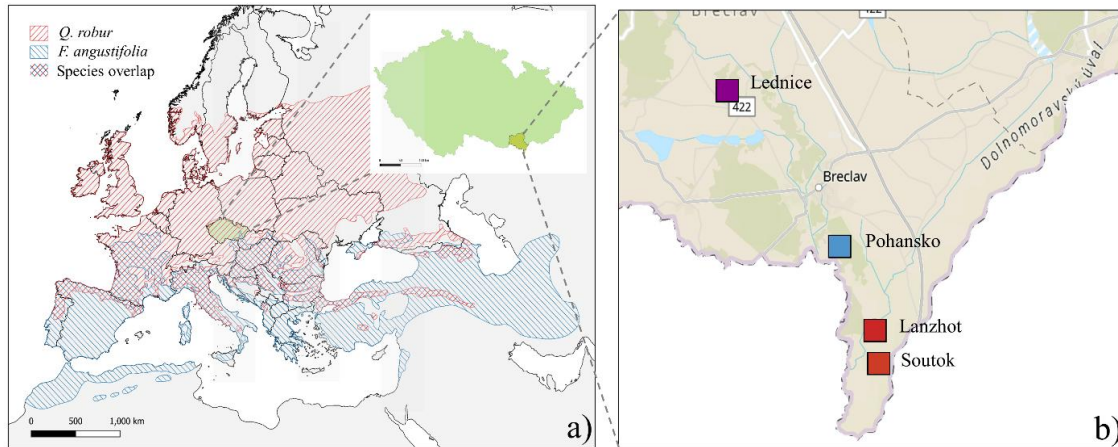


Figure 3. Overview of the study area and climate: (a) Map of Europe showing the natural ranges of oak (red) and ash (blue), with overlap areas shaded (Caudullo et al., 2017). The inset highlights the location of the study region in the Czech Republic. (b) Detailed map of the study area showing the four research sites: Lednice, Pohansko, Lanzhot, and Soutok (Basu et al., 2024).

The site is situated on an alluvial plain, characterized by predominately eutric humic fluvisol, haplic fluvisol, and eutric fluvisol (FAO, 2014), with soil depths varying from 60 to 120 cm (Acosta et al., 2017; Klimo and Hager, 2001). The sampling sites comprise floodplain forest stands that are around 110–120 years old, consisting of typical hardwood species (Table S1). Janik. et al. (2008) classified these plant communities within the "drier wing" of the *Fraxino pannonicae–Ulmetum* association, particularly in the *Fraxino pannonicae–Ulmetum carpinetosum* subassociation. Oak and ash constitute the principal canopy of the stand, signifying the main and co-dominant social classes. The common hornbeam (*Carpinus betulus* L.) primarily forms the understory layer, whereas a substantial population of juvenile field maple (*Acer campestre* L.) is observed in both the Pohansko and Lednice.

Table 1: Summary of species composition, basal area percentages, stand age, and average tree dimensions (DBH and height) for oak, ash, and other broadleaved species across four floodplain forest sites in South Moravia, Czech Republic.

Site	Coordinates	Species	Basal Area (%)	Stand Age (years)	Mean DBH (cm)	Mean Height (m)
Lednice	(48.80958, 16.78625)	Oak	51.7	120	68.995	34.26
		Ash	29.5		73.006	33.53
		Other broadleaved species	18.8			
Pohansko	(48.72694, 16.90611)	Oak	36.8	120	73.654	31.27
		Ash	55.9		73.005	32.64
		Other broadleaved species	7.3			
Soutok	(48.66346, 16.95202)	Oak	46.8	120	64.769	30
		Ash	31.5		72.2	26.33
		Other broadleaved species	21.7			
Lanzhot	(48.68161, 16.94597)	Oak	25.4	110	50.76	30.61
		Ash	39.9		57.57	35.93
		Other broadleaved species	34.7			

3.1.2 European oak network

The locations extend from 11.2°E to 23.4°E longitude and 44.5°N to 54.4°N latitude, which includes a range of climatic conditions throughout a broad latitudinal gradient and differing heights. All locations fit within the native distribution range of oak (Fig. 4).

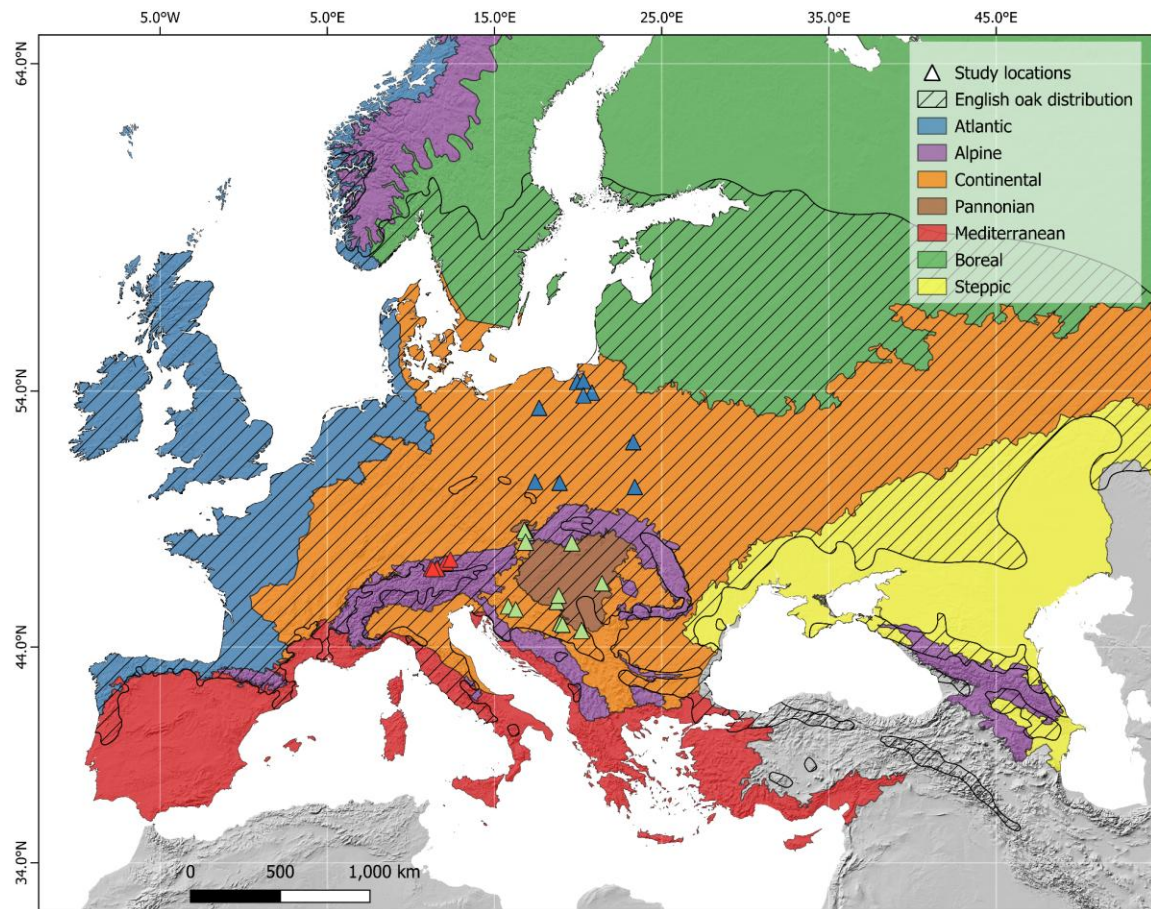


Figure 4. Overview of study area locations. Geographical distribution of the 33 study sites within the natural distribution range of oak (Caudullo et al., 2017) and within the biogeographic regions of Europe (Metzger et al., 2005). Adopted from (Stojanović et al., 2025).

The k-means clustering approach was employed to categorize locations according to mean annual temperature (MAT), resulting in the identification of three separate clusters: north, south, and high (Fig. 5).

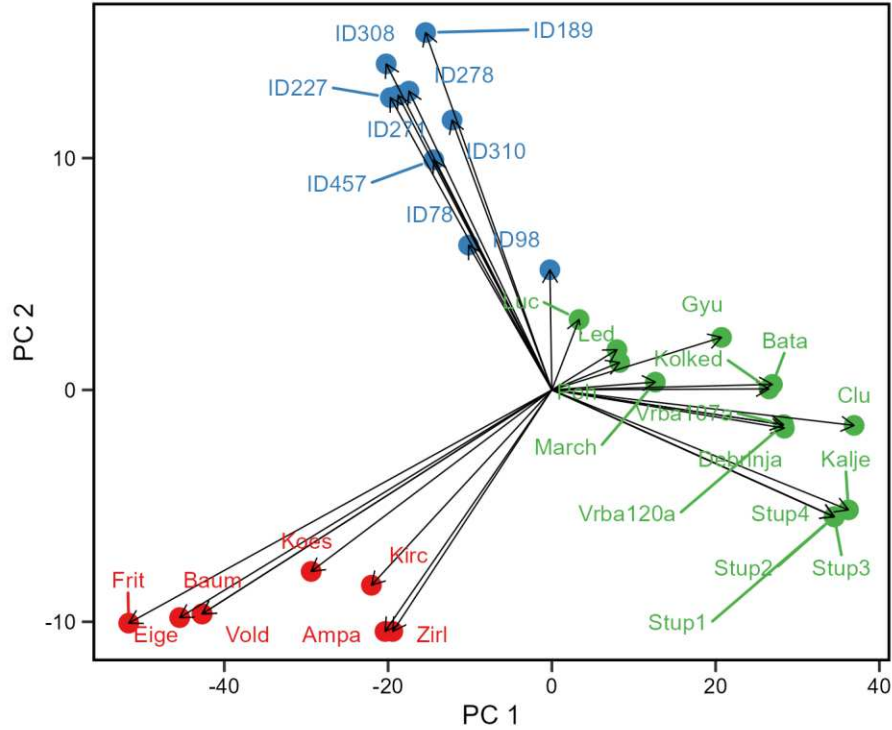


Figure 5. K-means clustering of the study sites based on mean annual temperature (MAT) resulting in three distinct clusters: high (red), north (blue) and south (green). Adopted from (Stojanović et al., 2025).

3.2 Groundwater and climate data:

3.2.1 Czech Republic:

For each location, we acquired data on daily mean, minimum, and maximum temperatures, together with daily precipitation totals, from the nearest grid point in the E-OBS data version 25.e (Cornes et al., 2018). The study region has experienced a significant long-term rise in temperature (Fig. 6; $p < 0.001$). In contrast, precipitation showed no significant long-term trend, with only a slight increase in summer rainfall. We utilized maximum and minimum temperatures to calculate potential evapotranspiration using the Hargreaves-Samani method (Hargreaves and Samani, 1985), essential for computing the SPEI developed by Vicente-Serrano et al. (2010). SPEI fluctuates between negative and positive values, indicating dry and wet seasons, respectively, and offers a more thorough perspective on actual plant water availability. This is accomplished by calculating

cumulative climatic water balances (Thornthwaite, 1948), considering the influences of both precipitation and temperature on potential evapotranspiration (PET). The SPEI values for the study sites were calculated monthly, spanning from 1960 to 2017, and aggregated over scales ranging from 1 to 12 months. The SPEI was computed utilizing the “SPEI” R package (Beguería et al., 2014).

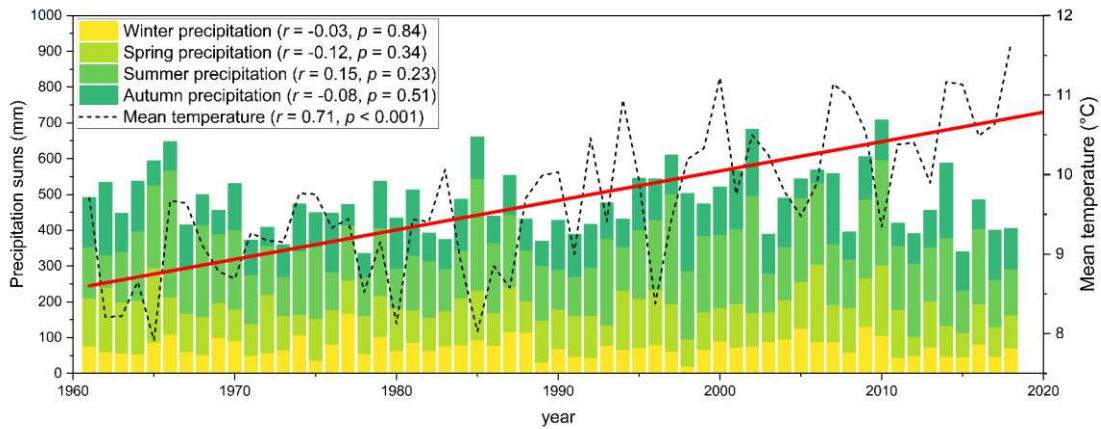


Figure 6. Climate diagram presenting long-term regional precipitation data (stacked columns for seasonal precipitation) and mean annual temperature (dashed line) with corresponding linear trend. Adopted from (Basu et al., 2024).

For Czech sites, groundwater levels have been recorded weekly by the Czech Hydrometeorological Institute since 1968 in Pohansko, 1958 in Lednice, and since 1967 in Soutok and Lanzhot, which utilize the same monitoring site. Weekly measurements were consolidated into monthly data and subsequently utilized in statistical analysis. The study period was restricted from 1960, due to lower quality of time series before. Groundwater levels showed different trends across the study sites (Fig. 7). In Lednice, levels remained stable with no significant change ($p = 0.18$). The average depth was 2.89 m below the surface. In Pohansko, groundwater levels dropped significantly ($p < 0.001$). The depth dropped from 2.2 m in 1960 to 3.3 m from surface in 2018. At Lanzhot and Soutok, near the Morava and Thaya rivers, the initial water table was shallowest at 1.6 m. After river modifications, a significant decline occurred over the study period ($p < 0.01$; Fig. 7)

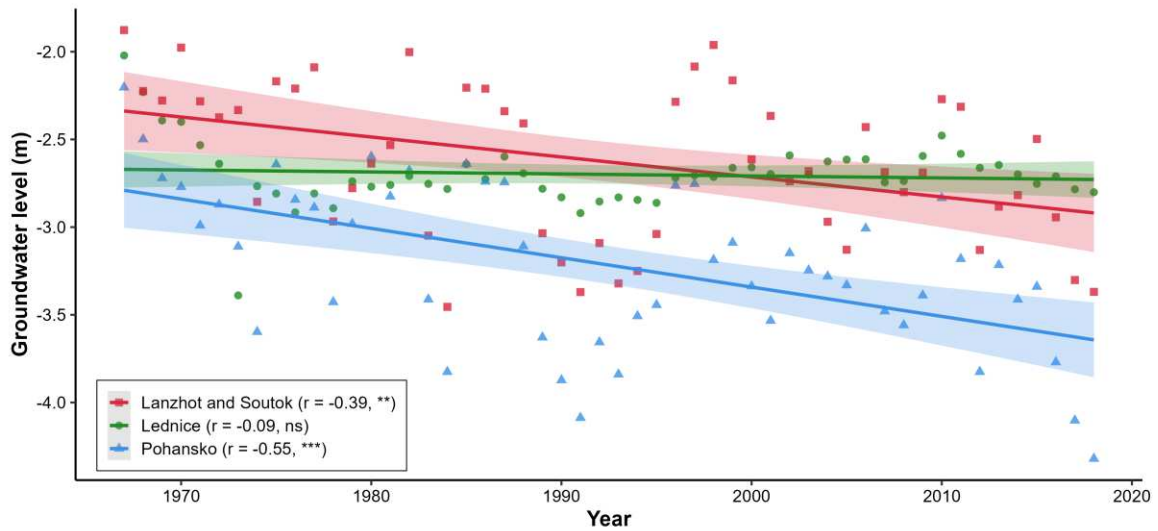


Figure 7. Long-term annual groundwater levels for each plot, including linear trends with indicated 95 % confidence interval in shaded color. Pearson's correlation coefficients (r) indicating the trends and their significance are provided in the legend of the figures; $p < 0.001$ (***), $p < 0.01$ (**), $p < 0.05$ (*), non-significant (ns). Adopted from (Basu et al., 2024).

3.2.2 European oak network

The "north" group predominantly occurs in the Continental bioregion of Poland, characterized by a MAT of 7.7°C and MAP of 607.1 mm (Fig. 8). The "south" group include the Pannonian bioregion spanning the Czech Republic, Slovakia, Hungary, Serbia, and the Continental bioregion of Croatia, characterized by a mean annual temperature of 11.2°C and an annual precipitation of 680.3 mm.

Table 2: General description of study sites and tree-ring chronologies. Column 'Group' denotes the site's cluster membership as determined by climate-based grouping analysis.

Code	Country	Group	Latitude	Longitude	Altitude	MAT	MAP
Ampa	Austria	High	47.271	11.483	574	7.2	960.65
Baum	Austria	High	47.295	11.550	593	5.2	1067.40
Eige	Austria	High	47.283	11.196	637	4.9	1089.42
Frit	Austria	High	47.306	11.600	573	4.4	1218.38
Kirc	Austria	High	47.524	12.077	490	7	1520.38
Koes	Austria	High	47.646	12.364	663	6.4	1734.31
Vold	Austria	High	47.284	11.558	570	5.2	1067.40
Zirl	Austria	High	47.267	11.304	691	7.2	987.17
ID189	Poland	North	52.166	23.298	158	7.6	542.76
ID227	Poland	North	50.491	23.389	263	7.3	628.15
ID271	Poland	North	54.356	20.318	143	7.2	665.57
ID278	Poland	North	53.849	20.311	175	7.4	617.97
ID308	Poland	North	53.949	20.826	163	7.1	616.17
ID310	Poland	North	54.320	19.936	57	7.8	624.95
ID457	Poland	North	53.406	17.669	131	7.6	534.06
ID78	Poland	North	50.651	18.892	316	8.1	658.62
ID98	Poland	North	50.684	17.424	172	8.9	575.81
Bata	Hungary	South	46.178	18.836	100	11	522.60
Clu	Serbia	South	44.713	20.205	73	12	628.87
Debrinja	Croatia	South	44.957	19.067	93	11	676.60
Gyu	Hungary	South	46.681	21.424	80	11	472.00
Kalje	Croatia	South	45.599	16.270	107	12	899.97
Kolked	Hungary	South	45.911	18.727	95	11	554.24
Led	Czechia	South	48.810	16.786	162	9.7	484.43
Luc	Slovakia	South	48.311	19.624	230	9.4	569.30
March	Austria	South	48.342	16.858	162	10	509.87
Poh	Czechia	South	48.727	16.906	164	9.7	499.37
Stup1	Croatia	South	45.730	15.814	141	12	935.54
Stup2	Croatia	South	45.729	15.814	141	12	935.54
Stup3	Croatia	South	45.729	15.814	141	12	935.54
Stup4	Croatia	South	45.729	15.814	141	12	935.54
Vrba107a	Croatia	South	45.024	18.921	94	11	662.51
Vrba120a	Croatia	South	45.017	18.932	86	11	662.51

The "high" group is found at mid-elevations in the Alpine bioregion of western Austria, characterized by humid weather, a MAT of 5.9°C, and a mean annual precipitation (MAP) of 1205.6 mm.

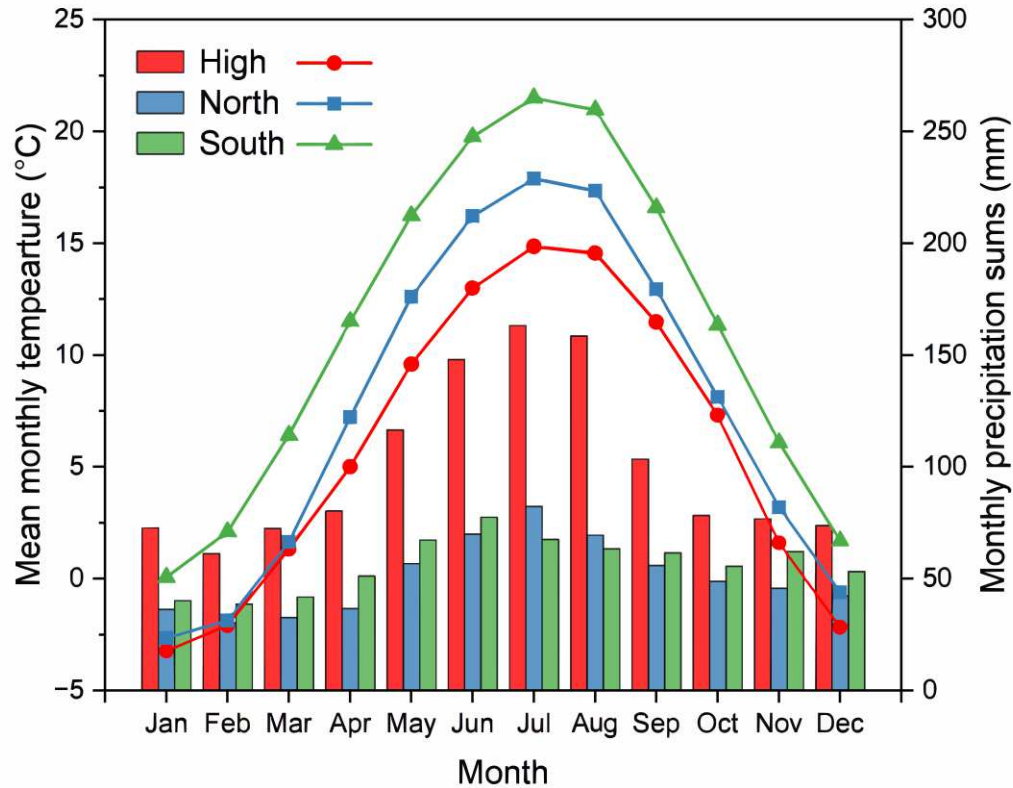


Figure 8. Climatic diagram showing the mean monthly temperature (solid lines) and monthly precipitation sums (columns) for the three observed clusters over the 1950–2022 period, based on E-OBS data (Cornes et al., 2018). Adopted from (Stojanović et al., 2025).

3.3 Dendroecological sampling:

3.3.1 Tree ring

Two increment cores (5 mm diameter) were obtained from dominant and co-dominant oak and ash trees at breast height (1.3 m) on opposing sides of the trunks between 2017 and 2021 in the Czech Republic. In total samples from 60 oak and 62 ash trees were used for final study in Czech Republic. Furthermore, for broader oak gradient study, we included

an extensive network comprising 33 oak-dominated sites, with a total of 773 oak trees and approximately 66,000 ring-width measurements.

All cores were air-dried, mounted, and successively sanded with finer sandpaper up to 600 grit until the tree rings were clearly visible (Stokes and Smiley, 1968), then measured to 0.01 mm accuracy with a stereoscopic microscope and Lintab measuring table. Using PAST5TM software (SCIEM), tree-ring measurements were visually and statistically cross-dated and synchronised. On-screen comparisons and statistical parameters like the Gleichläufigkeit coefficient (Buras and Wilmking, 2015; Eckstein and Bauch, 1969) and the Baillie-Pilcher t-value (Baillie and Pilcher, 1973) were used, and COFECHA (Holmes, 1983) was used for additional quality checks. To eliminate non-climatic growth trends, raw data series were standardised and detrended for each tree utilising a cubic spline with a 50% frequency response at 32 years (Klesse et al., 2016); the detrended chronologies were subsequently pre-whitened and averaged employing a robust bi-weight mean to generate site chronologies. The reliability and quality of these chronologies were evaluated by computing various dendrochronological statistics for the period 1920–2005: first-order autocorrelation (Ar_1) of raw widths (Fritts, 1976; Speer, 2010) mean sensitivity (MS) of indexed series, mean interseries correlation (R_{bar}), signal-to-noise ratio (SNR) and the expressed population signal (EPS), with EPS values ≥ 0.85 deemed adequate for climate–growth analysis (Buras, 2017; Wigley et al., 1984). Detrending and other statistical analyses were conducted utilising the “dplR” package (Bunn, 2010).

3.3.2 Anatomy study

At each site, we selected four to five robust individuals per species representing the dominant or co-dominant canopy layer for wood anatomical analysis (Bryukhanova and Fonti, 2013; Pipíšková et al., 2025). Notably, these individuals were distinct from those used in the previous tree-ring width study, ensuring independent datasets for anatomical and dendrochronological analyses. Increment cores (5 mm diameter) were extracted at approximately 1.3 m above ground from both the northern and southern aspects of each tree using a Pressler borer, to account for potential radial growth differences. All cores were stored in low-humidity conditions to prevent mold growth prior to laboratory analysis. For anatomical preparation, we used a WSL Sledge Microtome (Gärtner et al., 2015) to

produce smooth, flat surfaces for microscopic imaging while preserving anatomical features such as vessels and fibers. High-resolution images were acquired using a Zeiss LSM 800 laser scanning confocal microscope (Plan-Apochromat 5×/0.16 objective) with a 488 nm excitation laser, bidirectional scanning (pixel dwell time 2.46 μ s), a confocal pinhole of 1 AU (36 μ m), and a PMT detector gain of 800 V, resulting in pixel dimensions of 744 $\times$ 744 nm. To capture the full sample area, the Tiles module was used to merge 31 tiles per sample. Images were processed in FIJI (Schindelin et al., 2012) for contrast enhancement and segmentation, using automated thresholding and manual refinement.

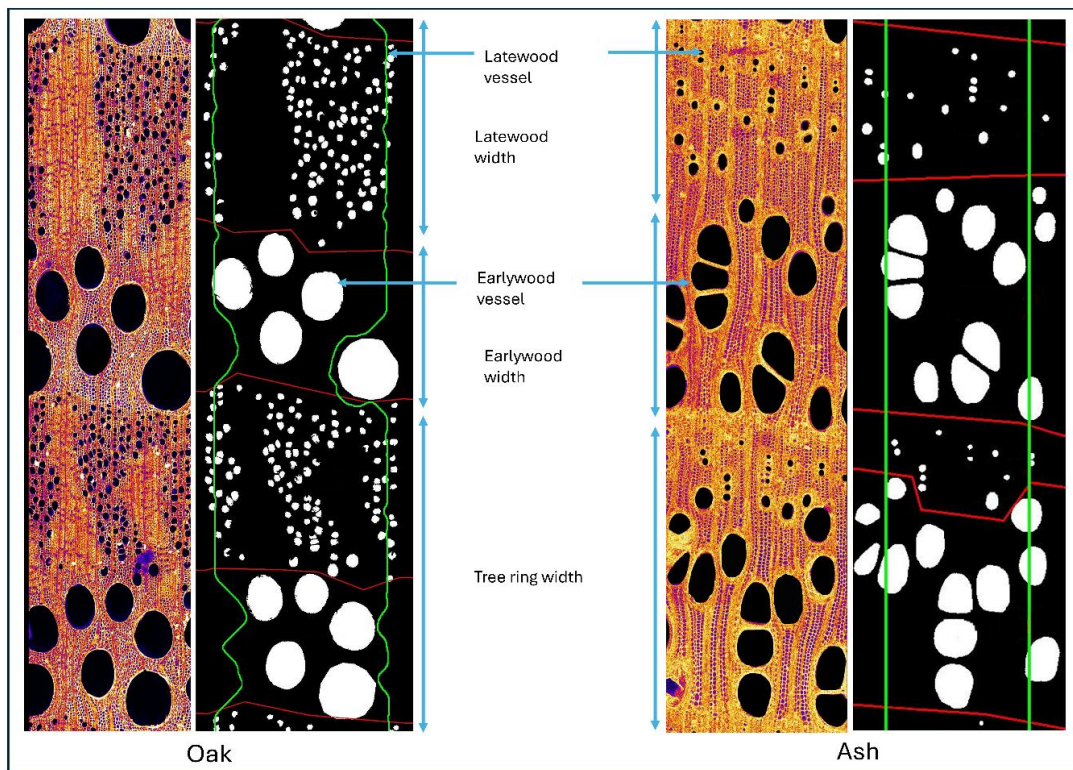


Figure 9. The figure presents cross-sectional scans of oak and ash tree ring, highlighting earlywood and latewood vessels. The black-and-white images depict automatically identified vessels using Roxas, where the vessels are shown as white against a black background. Tree-ring width, earlywood width, and latewood width are indicated, with red boundaries

Wood anatomical parameters for earlywood and latewood were measured for each year from 1970 to 2023 using ROXAS software (Fig. 9, von Arx and Carrer, 2014). We quantified mean lumen area (MLA) the average cross-sectional area of the vessel lumens

along with vessel density (CD), defined as the mean number of water-conducting vessels per mm². The hydraulically weighted vessel diameter (Dh), which emphasizes the contribution of larger vessels to water transport, was calculated as $Dh = (\sum D^4/N)^{(1/4)}$, where D is vessel diameter and N is the number of vessels (Tyree and Zimmermann, 2002). Theoretical hydraulic conductivity (Kh), representing the potential efficiency of water movement through wood, was estimated using the Hagen–Poiseuille equation, $Kh = \pi D^4/128\eta$, where η is the viscosity of water (1.002×10^{-9} MPa·s at 20°C), and is expressed in [m⁴ MPa⁻¹ s⁻¹].

We used Generalized Additive Mixed Models (GAMM) to investigate the effects of climate, tree size, and legacy on earlywood MLA in ash and oak at the individual tree level, employing the “*mgcv*” package in R (Wood, 2011). The response was the current year’s earlywood MLA, with the previous year’s latewood MLA included as a predictor to capture legacy effects. Tree size was represented by the calculated DBH for each year, derived from the DBH measured during sampling and adjusted backward using annual tree-ring widths, then log-transformed to obtain normality. Monthly climate variables (precipitation, groundwater level, and temperature from February to July) were tested individually as smooth terms to capture nonlinear responses. Location was fitted as a fixed effect, and random effects were included for tree identity and year to account for repeated measurements and temporal structure. Model selection was based on delta AIC to compare alternative model specifications and identify the best-fitting and simplest model. Predictor effects were assessed using F-values and significance levels, and partial effect plots were generated to visualize species-specific responses.

3.4 Tree ring - climate growth correlation

Climate-growth correlations for the Czech floodplain sites were investigated utilizing the *monthly_response()* function from the “*dendroTools*” R package (Jevšenak, 2020, 2019). Standardized tree-ring width indices (TRWi) were associated with monthly meteorological variables, including mean temperature (Tmean), precipitation, groundwater level, and the SPEI. The analysis encompassed both the current and preceding growing seasons, employing moving windows ranging from 1 to 12 months. To assess temporal stability in

climate sensitivity, the dataset was split into 20-year intervals (starting from 1961–1980 and ending in 1998–2017).

For the larger oak network, climate-growth correlations were determined for the period 1960–2022 utilizing TRWi and monthly climatic variables (Tmin, Tmax, precipitation, and SPEI). The analysis focused on the periods from July to October of the preceding year and from March to October of the current year, based on documented xylogenesis phenology and non-structural carbohydrate dynamics in ring-porous species (Michelot et al., 2012; Noyer et al., 2023). The maximum absolute correlation at each location ($r_{\max}$) was obtained to signify peak climatic sensitivity. We employed a multivariate linear regression model to explain geographical variance in $r_{\max}$, utilizing mean annual temperature (MAT), mean annual precipitation (MAP), and elevation as predictors:

$$r_{\max} = \beta_0 + \beta_1 \times MAT + \beta_2 \times MAP + \beta_3 \times Elevation + \varepsilon$$

Model selection was carried out by a comprehensive subset assessment utilizing the Bayesian Information Criterion (BIC) to identify the most parsimonious predictor combination. The resultant model was subsequently employed to spatially interpolate $r_{\max}$ throughout the species' native range, clarifying geographic patterns of oak climate sensitivity.

3.5 Disturbance year identification and tree-ring based resilience analysis

3.5.1 Identification of disturbance years

To assess tree growth reactions to severe environmental stress, disturbance years were determined by two distinct methodologies, customized for the scale and data context of each investigation.

3.5.1.1 Selection of drought years in Czech Republic using SPEI

At the South Moravian floodplain locations (Lednice, Pohansko, Lanzhot, Soutok), disturbance years were determined according to hydrometeorological drought conditions. We employed SPEI-12, computed from the preceding October to the current September,

to represent the complete hydrological year. The three years with the lowest SPEI-12 values from 1960 to 2014 were identified as representative extreme dry events (Fig. 10). This approach ensures alignment with long-term meteorological data and identifies physiologically significant dry intervals that are likely to influence radial growth.

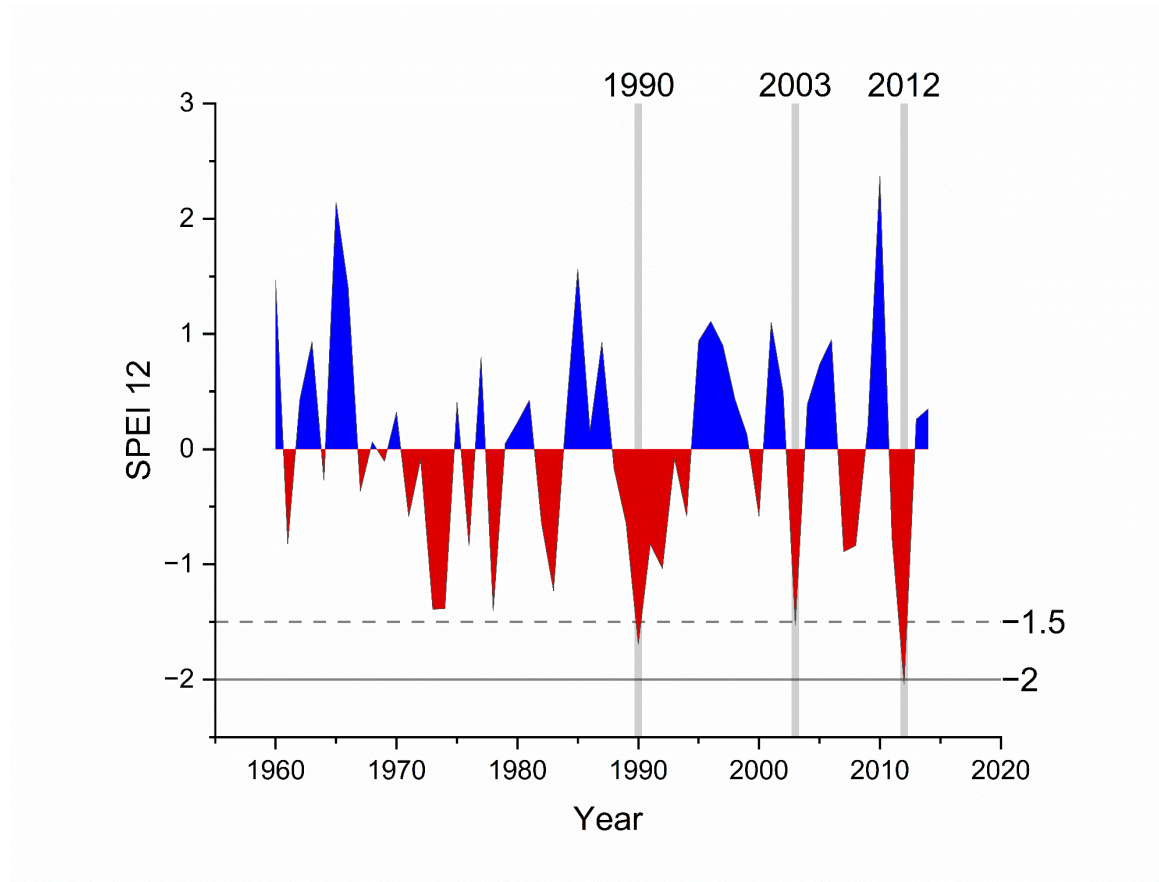


Figure 10. The 12-month Standardized Precipitation Evapotranspiration Index (SPEI 12), covering October of the previous year to September of the current year from 1960 to 2014, is shown. Dashed and solid gray lines indicate very dry and extremely dry conditions, respectively. Vertical gray bands highlight the identified pointer years 1990, 2003, and 2012, which were subsequently used to calculate resilience components. Adopted from (Basu et al., 2024).

3.5.1.2 European oak network: Year of pointer detection

For the broader oak network across nine European countries along a temperature gradient, disturbance years were identified using a pointer-year approach. Negative pointer years indicate significant declines in radial development, most likely resulting from drought, late-spring frost, insect defoliation, masting, or various biotic and abiotic causes (Bräuning et

al., 2016; Vitasse et al., 2019). To enhance robustness, we utilized five detection algorithms incorporated in the “pointRes” R package (Maaten-Theunissen et al., 2021). A year was designated as a negative pointer year if it was recognized by a minimum of three out of five techniques. This methodology enabled the identification of a wider array of growth-restricting events, enhancing comparability across ecological gradients.

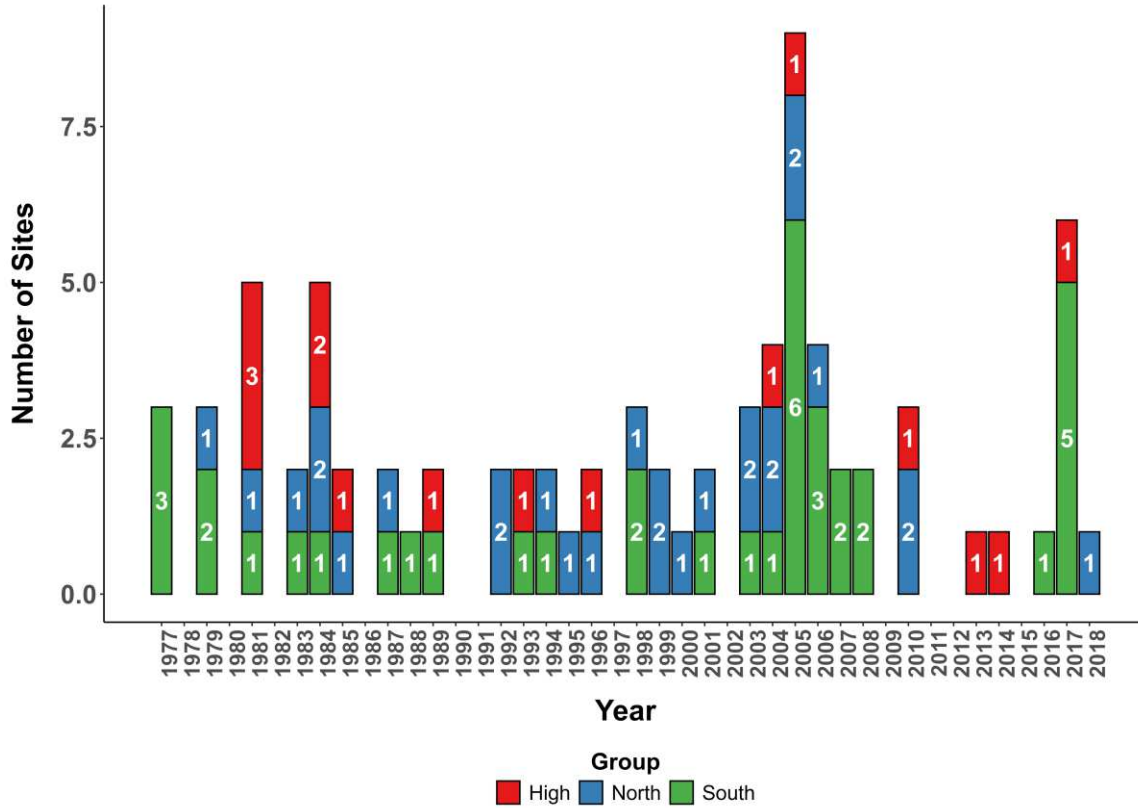


Figure 11. Number of sites recording negative pointer years from 1977 to 2018, categorized by biogeographical clusters: High, North, and South (Stojanović et al., 2025).

3.5.2 Calculation of resilience components

For each identified disturbance year, we computed three tree-ring-based resilience indices to quantify different aspects of growth response and recovery following stress (Lloret et al., 2011):

$$R_t = \frac{RW}{pre\ RW}$$

Resistance (R_t) quantifies the extent to which tree growth is reduced during the disturbance year relative to the mean growth in the three preceding years (*pre RW*). High resistance indicates better immediate tolerance to stress (e.g., drought or defoliation), whereas low values reflect greater initial impact.

$$R_c = \frac{Post\ RW}{RW}$$

Recovery (R_c) captures the tree's ability to restore growth after the disturbance. Here, *Post RW* is the mean growth in the three years following the event. High R_c values indicate rapid physiological rebound and reactivation of cambial activity and resource allocation.

$$R_s = \frac{Post\ RW}{Pre\ RW}$$

Resilience (R_s) reflects the tree's capacity to return to pre-disturbance growth levels. It integrates both suppression and recovery, providing a comprehensive view of post-event stability. Values near or above 1 imply full recovery or even compensatory growth; values below 1 indicate lasting suppression.

Additionally, we applied the “full resilience” framework proposed by Schwarz et al. (2020) to benchmark the resilience of all three clusters against a hypothetical line of full resilience. The line of full resilience was calculated as follows

$$R_c = \frac{R_s}{R_T} = \frac{1}{R_T} = R_T^{-1}$$

where resilience is fixed at a constant value of 1, denoting complete recovery capacity of trees, regardless of resistance (R_t). Subsequently, the Nash–Sutcliffe efficiency (NSE; Nash and Sutcliffe (1970)) was used to evaluate the similarity between observed and a hypothetical line of full resilience across the three oak clusters. NSE quantifies how well the observed versus hypothetical values conform to a 1:1 full-resilience line, providing a measure of model performance (Krejza et al., 2017). It was calculated using

$$NSE = 1 - \left[\frac{\sum_{i=1}^n (Y_i^{obs} - Y_i^{hyp})^2}{\sum_{i=1}^n (Y_i^{obs} - Y^{obs.mean})^2} \right]$$

All indices were calculated using the “pointRes” R package (Maaten-Theunissen et al., 2021). The analysis was performed at the individual-tree level using raw (non-detrended) tree-ring width data to retain absolute growth magnitude. A four-year window was used for both pre- and post-disturbance periods, representing short-term physiological dynamics without overlapping unrelated growth trends.

This framework enables consistent cross-site and cross-species comparison of growth performance and resilience under disturbance, supporting the broader objective of identifying vulnerability and adaptation patterns in floodplain and lowland forests.

3.6 Use of Artificial Intelligence

In accordance with the 2023 statement ‘Zásady využívání umělé inteligence na Mendelově univerzitě v Brně’, OpenAI’s generative artificial intelligence ChatGPT (model GPT-4.5) was used solely for language correction, improvement of readability; all scientific content, analyses, and interpretations remain the author’s own.

4. Results

4.1 Local-scale responses in Czech floodplain forests

4.1.1 Tree-ring analysis of oak and ash

4.1.1.1 Growth patterns and long-term trends

At all study sites, ash and oak together accounted for more than 60% of the total stand basal area (Table. 1). These two species form the structural and ecological most important of the floodplain forests. Tree-ring analysis revealed clear differences in their growth behaviour. Oak showed higher first-order autocorrelation values, ranging from 0.69 to 0.75, whereas ash ranged from 0.61 to 0.64 (Table 1). This suggests oak growth is more strongly influenced by the previous year's conditions, indicating greater persistence in its annual growth pattern.

These values reflect a comparable level of inter-annual variability and environmental responsiveness. However, oak exhibited stronger synchrony among individual trees. Its R_{bar} was consistently higher between 0.52 and 0.55 than that of ash, which ranged from 0.46 to 0.52 (Table. 3). This points to a more coherent growth signal in oak, possibly due to a more uniform response to regional climatic factors.

Crucially, the EPS exceeded 0.91 at all sites for both species. This confirms the high reliability of the chronologies and their suitability for climate–growth analyses. While both species are valuable for dendroecological studies, oak offers a slightly more stable and consistent signal across the landscape.

Table 3: Overview of the basic quality assessment for the developed tree-ring chronologies, including site location, tree species, number of sampled trees (N), mean age, time span, first-order autocorrelation ($Ar1$), mean inter-series correlation (R_{bar}), and expressed population signal (EPS).

Location	Species	N	Mean age	Time span	$Ar1$	R_{bar}	EPS
Lanzhot	Oak	17	100	1906 – 2017	0.75	0.55	0.95

Lanzhot	Ash	19	102	1897 – 2017	0.64	0.52	0.95
Soutok	Oak	14	130	1876 – 2020	0.69	0.55	0.95
Soutok	Ash	16	124	1877 – 2020	0.66	0.46	0.93
Pohansko	Oak	13	122	1890 – 2019	0.79	0.52	0.93
Pohansko	Ash	14	105	1890 – 2020	0.61	0.49	0.93
Lednice	Oak	16	135	1879 – 2020	0.72	0.54	0.95
Lednice	Ash	13	116	1892 – 2020	0.62	0.46	0.91

Distinct differences in tree-ring width patterns between oak and ash emerged across the analysed sites (Fig 12a). The first two principal components from the PCGA explained between 56% and 57% of the total variance at all four locations (Fig. 12b). Pohansko and Lanzhot showed similar loading structures, with oak consistently positioned above ash. In contrast, Soutok presented the reverse pattern, where ash had stronger loadings than oak. Lednice stood out. Here, ash had lower loadings on PC1 but higher on PC2, forming a near-orthogonal orientation compared to the other sites, where PC1 dominated.

Across Pohansko, Soutok, and Lanzhot, oak and ash growth patterns were more aligned with one another, though still clearly distinct. In Lednice, however, the separation was more pronounced. A Wilcoxon signed-rank test confirmed statistically significant differences ($p < 0.05$) in the polar angles of the PCGA ranks between oak and ash at each site. The PCGA plots also showed well-defined groupings, reinforcing the differentiation in species-specific growth responses.

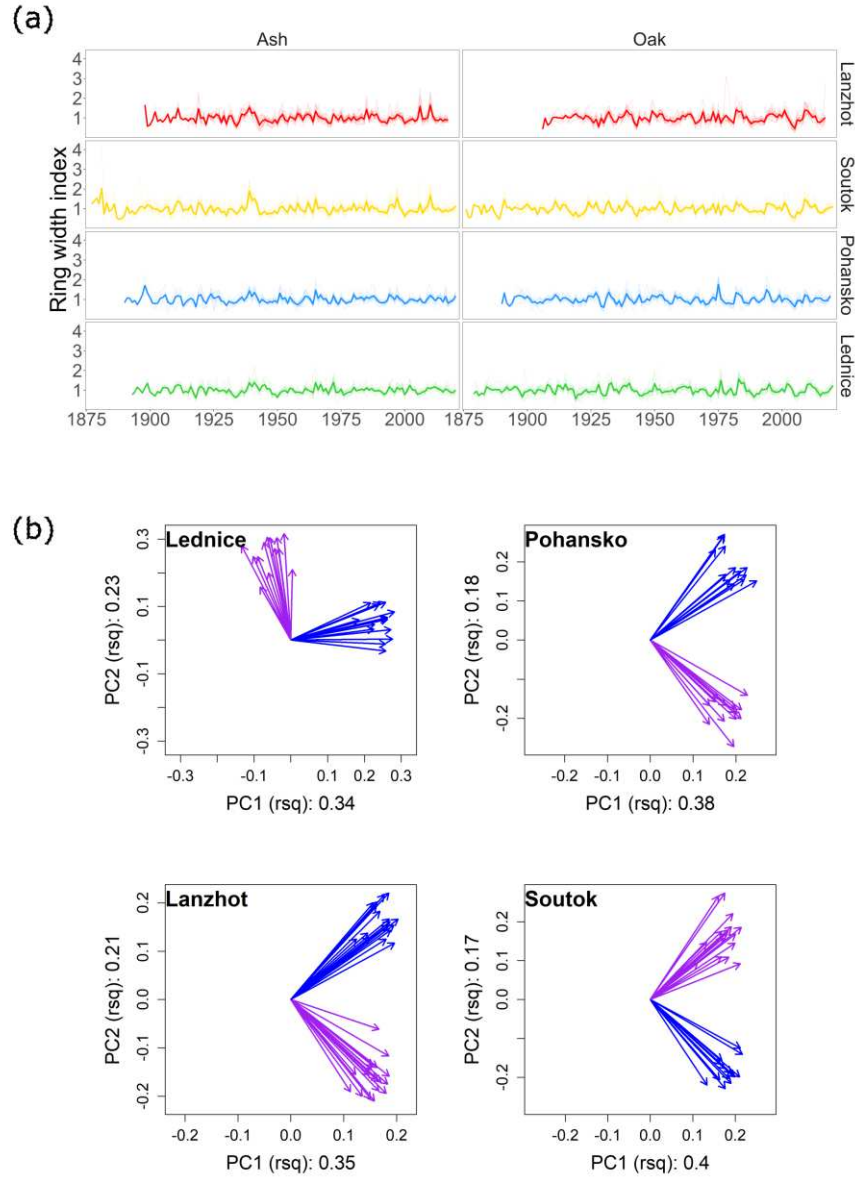


Figure 12. (a) Detrended ring-width chronologies for ash and oak across four floodplain forest sites: Lednice, Pohansko, Soutok, and Lanzhot. Each panel shows species-specific growth trends from 1875 to 2010, with ash on the left and oak on the right. (b) Principal Component Gradient Analysis (PCGA) of tree-ring data from the same four sites: Lednice, Pohansko, Soutok, and Lanzhot. Vectors represent individual trees, with oak populations shown in magenta and ash populations in blue, illustrating clear species-level separation along principal components. Adopted from (Basu et al., 2024).

4.1.1.2 Static climate–growth relationships

Our study identified strong and statistically significant correlations between hydroclimatic variables and radial tree growth (Figs. 13,14). In contrast, the influence of temperature on radial growth was relatively minor (Fig. 13). Ash showed a strong positive relationship between radial growth and groundwater levels. At Lednice, although the positive correlation between ash growth and groundwater was consistent with other sites, the strength of this relationship was weaker (Fig. 14). Notably, the positive correlation spanned from May to the end of October in the current year.

Ash also showed a positive correlation with precipitation from May through October (Fig. 13). The strongest correlation at all sites occurred in May. Again, Lednice showed the weakest correlation with precipitation among the four sites. Oak generally showed no significant correlation with groundwater, except at Pohansko, where groundwater from the previous spring and summer showed a negative relationship with growth (Fig. 14). Overall, correlations between oak and hydroclimatic variables were much weaker, in sharp contrast to the strong positive associations observed in ash.

SPEI, calculated over 1- to 3-month periods, showed the strongest effect on radial growth (Fig. 14). Ash growth had a strong positive correlation with SPEI from May through the end of the year, with the highest correlation observed in spring ($r = 0.64$; $p < 0.001$). In Lednice, however, this correlation was weak, suggesting a different response than at the other sites. The highest correlation at Lednice was observed between May and October, with a similar pattern from January to September. In August, the weakest correlation between SPEI 3 and ash growth was recorded in Lednice ($r = 0.31$; $p < 0.05$). In contrast, oak generally showed no significant correlation with SPEI. In conclusion, stronger correlations were seen in Lanzhot and Soutok, where groundwater levels are declining. The weakest correlations were found in Lednice, where groundwater remains high and stable.

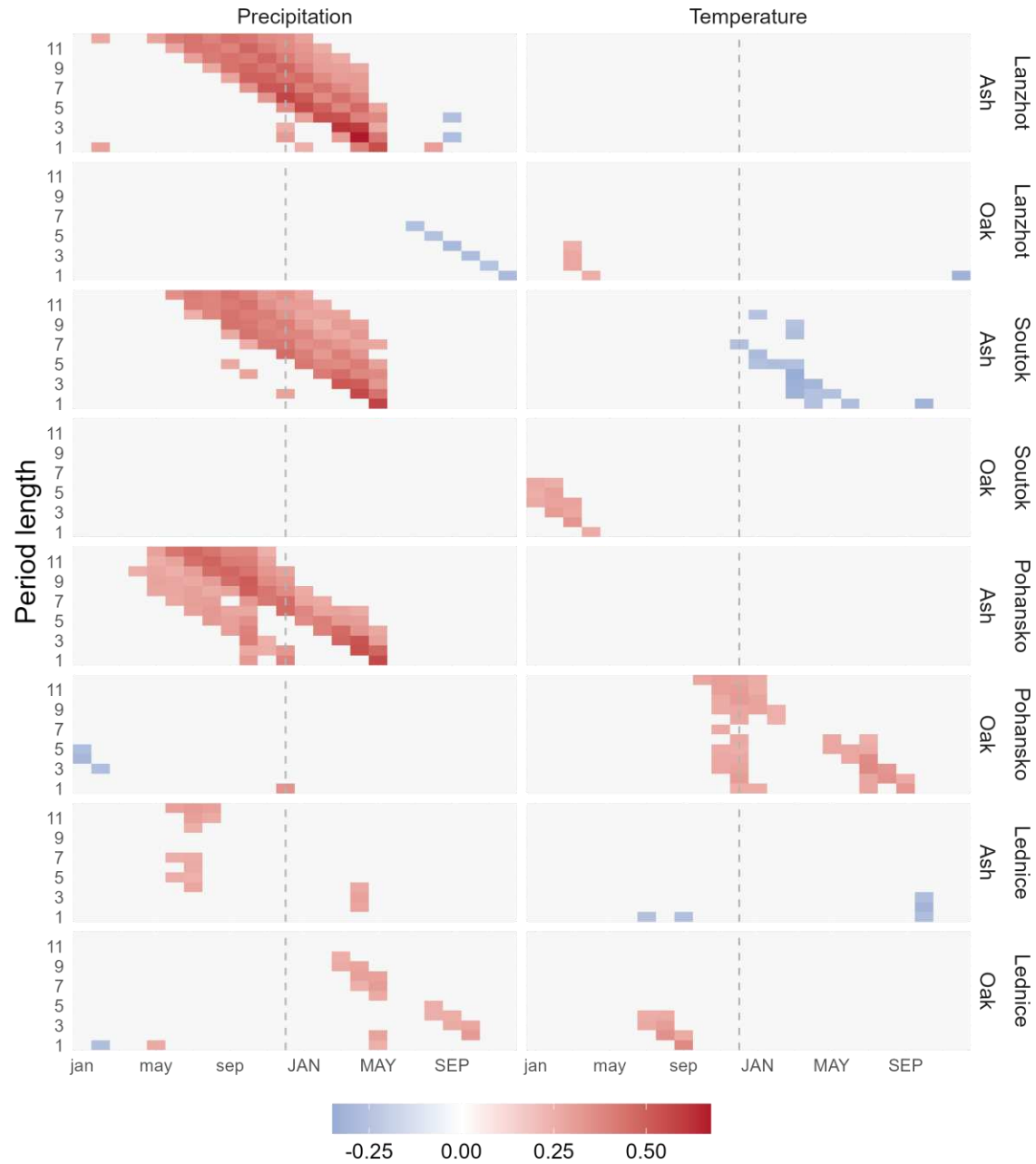


Figure 13. Pearson correlation coefficients between tree-ring indices and monthly precipitation and temperature. Only statistically significant correlations ($p < 0.05$) are displayed. The dashed vertical line marks the boundary between the previous and current calendar years. Adopted from (Basu et al., 2024).

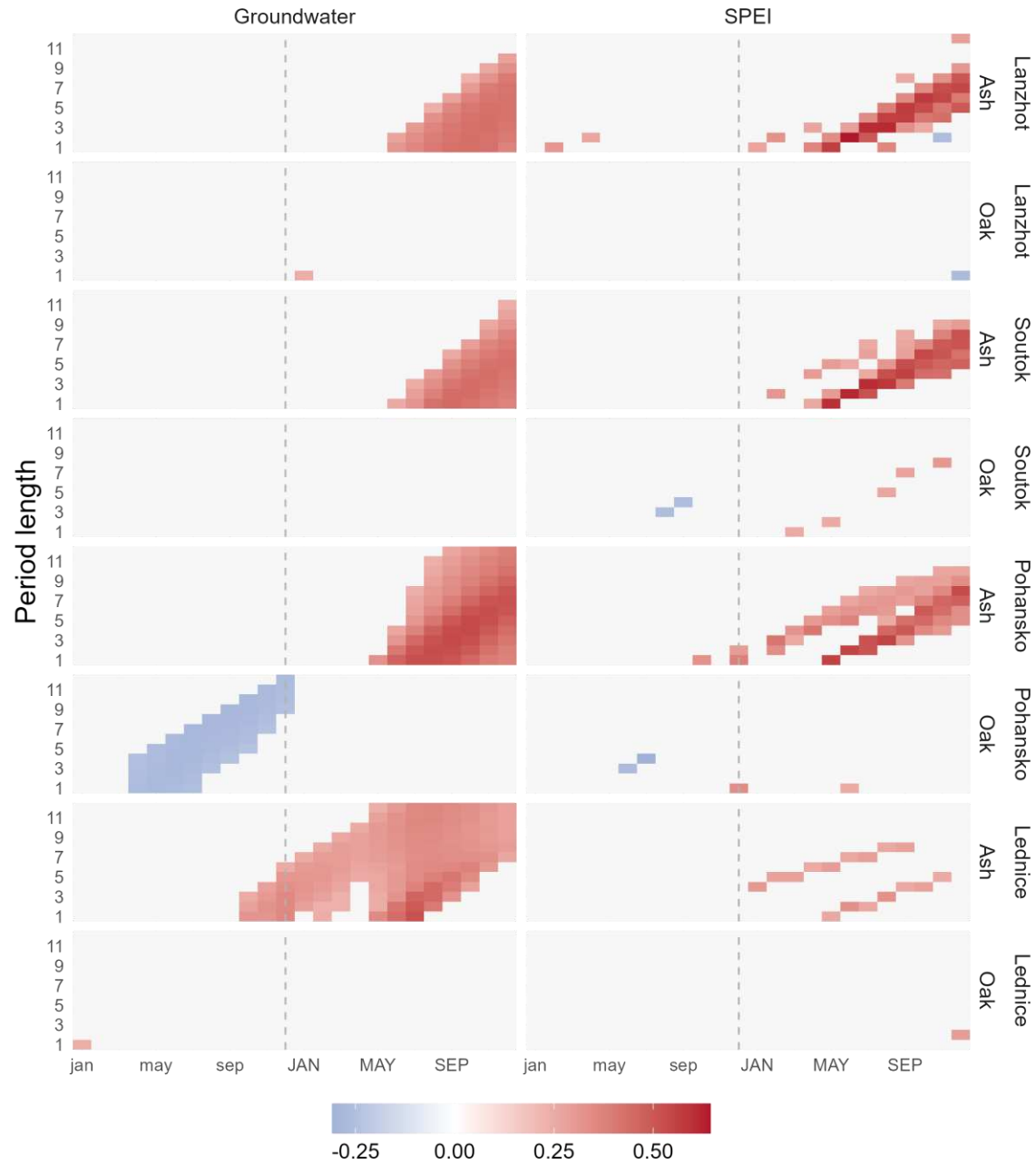


Figure 14. Pearson correlation coefficients between tree-ring indices and monthly groundwater and Standardized Precipitation-Evapotranspiration Index (SPEI). Only statistically significant correlations ($p < 0.05$) are displayed. The dashed vertical line marks the boundary between the previous and current calendar years. Adopted from (Basu et al., 2024).

4.1.1.3 Temporal shifts in climate sensitivity

Our moving-window analyses revealed clear differences in the temporal dynamics of correlations between hydroclimatic variables and the radial growth of ash and oak. Ash showed stable and consistent correlations with key hydroclimatic parameters, such as summer SPEI 3, May precipitation, and summer groundwater levels across several sites during the growing season (Figs. 15, 16). However, this pattern was less evident in Lednice, where ash appeared less dependent on these factors. Notably, the correlation between May precipitation and ash growth strengthened over time at most sites, with Lednice again being the exception.

In contrast, oak initially showed weaker correlations with hydroclimatic variables. Yet over time, the shift became apparent. At Lanzhot and Soutok, oak began to exhibit increasing sensitivity to these environmental factors. This suggests a developing trend in which oak is becoming more responsive to hydroclimatic conditions (Figs. 15, 16).

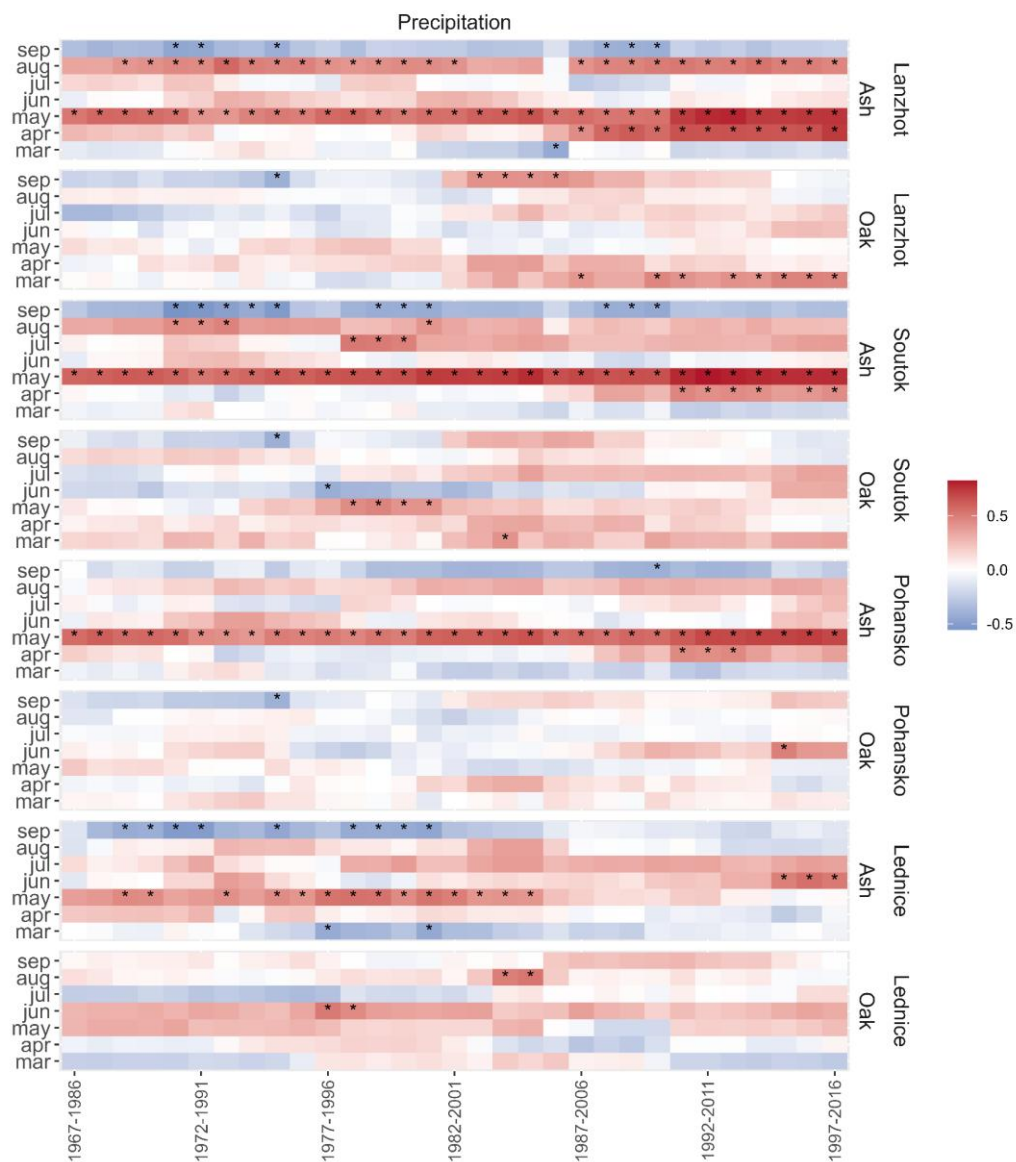


Figure 15. Moving correlations (20-year window) between ring width indices and precipitation for the period 1967–2018. Significant correlations with $p < 0.05$ are indicated by black asterisk. Adopted from (Basu et al., 2024).

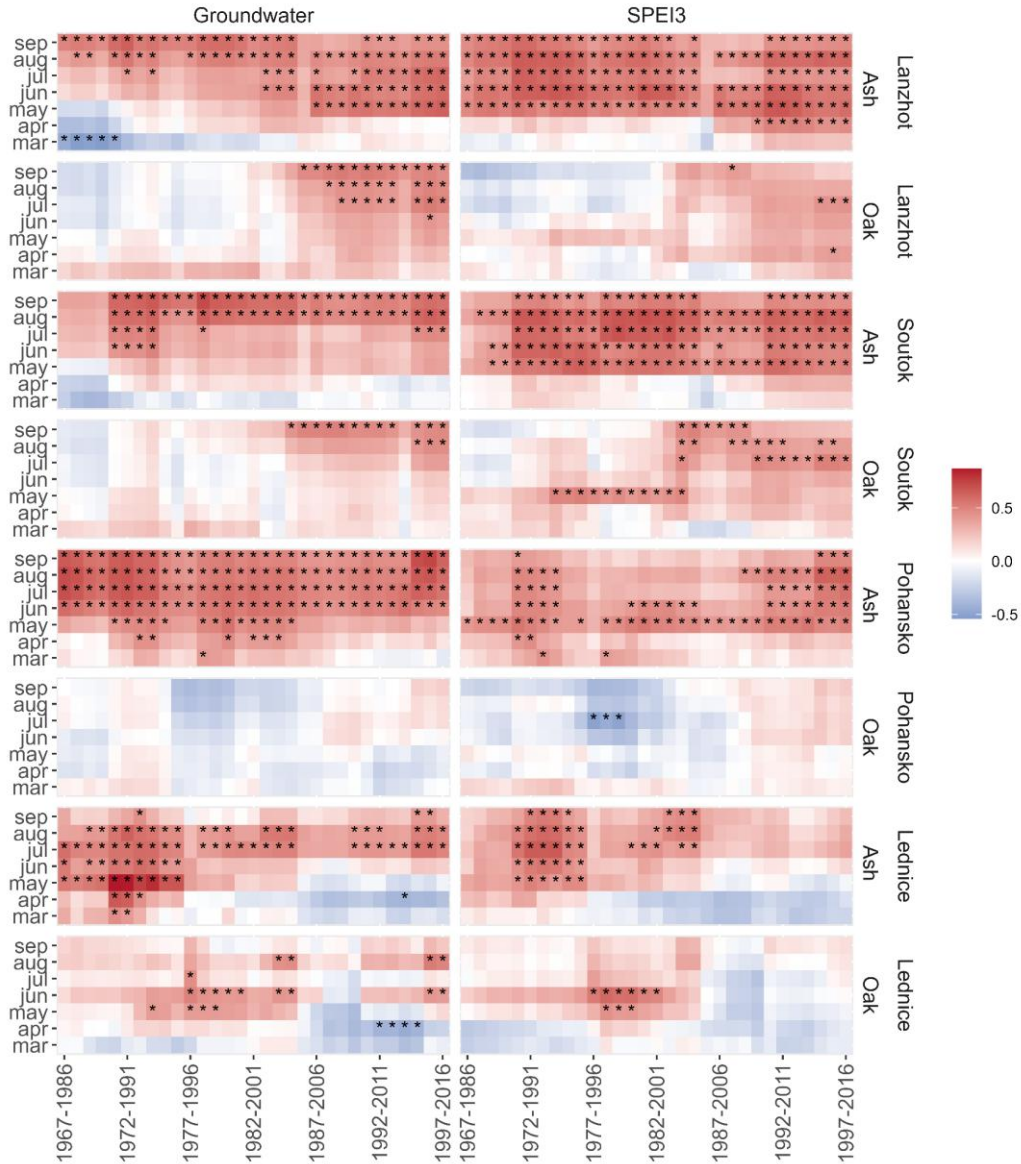


Figure 16. Moving correlations (20-year window) between ring width indices and groundwater and the 3-month Standardized Precipitation-Evapotranspiration Index (SPEI 3) for the period 1967–2018. Significant correlations with $p < 0.05$ are indicated by black asterisk. Adopted from (Basu et al., 2024).

4.1.1.4 Drought resilience during pointer years

An assessment of resilience during the driest years based on SPEI 12 specifically 1990, 2003, and 2012 (see Fig. 10) revealed clearly contrasting patterns. Among the four sites, oak in Lanzhot and Soutok consistently showed lower resilience and resistance across all drought years (Fig. 17). Oak in Lednice demonstrated reduced recovery compared to the other locations (Fig. 17).

Ash showed no clear spatial pattern in resilience, although most sites recorded values below 1. In Lednice, both species had resilience values below 1 during the drought years (Fig. 17). Regarding recovery, no consistent trend emerged across sites, but Lednice again performed differently; recovery was consistently below 1 for both oak and ash. Specifically, values for 2003 and 2012 were below 1 in both species. Conversely, Pohansko recorded higher resilience and recovery in 2012 than the other sites (Fig. 17).

When it came to resistance, oak generally had values close to 1 in 1990, while ash tended to be lower, except at Lednice. In 2003 and 2012, oak displayed similar resistance values across all four sites, though those in areas with declining groundwater levels, Lanzhot and Soutok tended to have lower resistance than those in Pohansko and Lednice. Of the three drought years, 2003 marked the lowest resilience for oak at all sites, while for ash, the lowest value occurred in 1990 at Lanzhot.

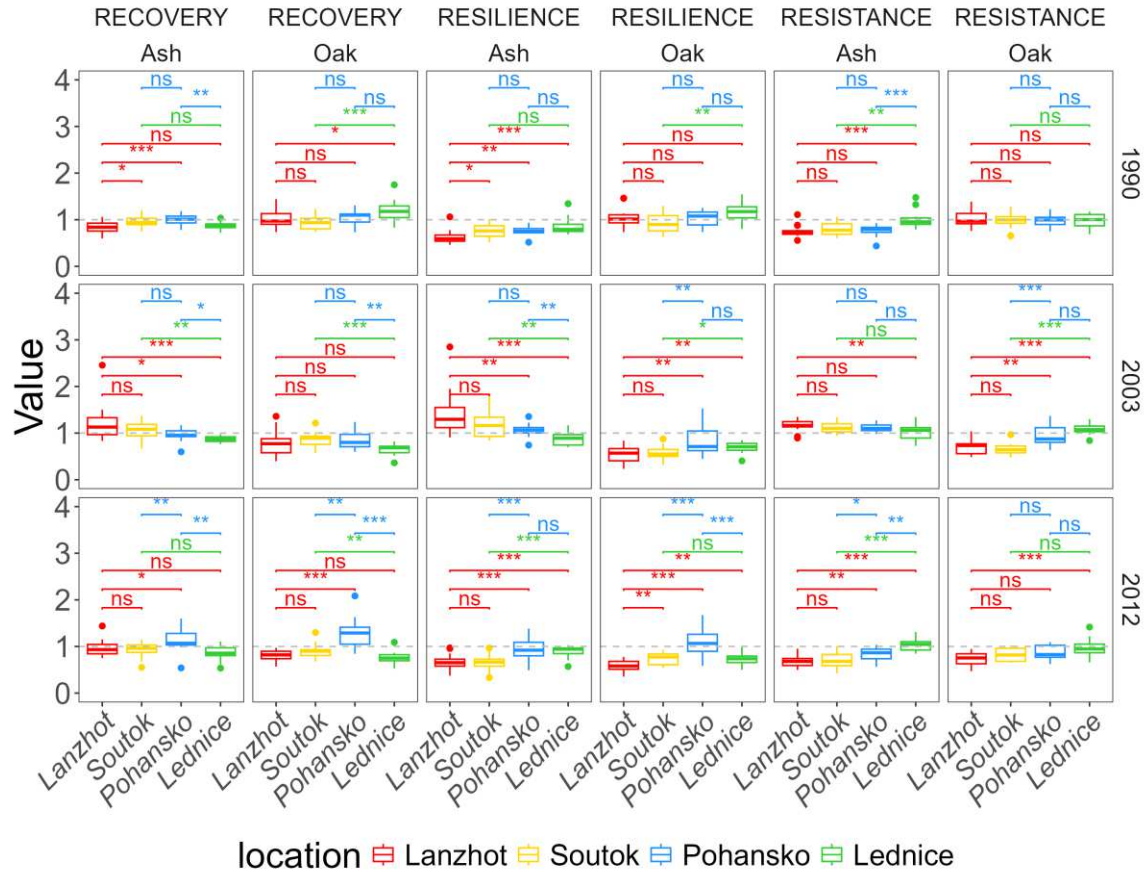


Figure 17. Recovery, resilience, and resistance indices for ash and oak during the driest years. Pairwise differences were assessed using the Wilcoxon signed-rank test. Significance levels: $p < 0.001$ (***), $p < 0.01$ (**), $p < 0.05$ (*), non-significant (NS.). Adopted from (Basu et al., 2024).

4.1.2 Wood anatomical responses in Czech floodplain forests

4.1.2.1 Anatomical variability in oak and ash

Hydraulic and anatomical traits varied by species and location, reflecting contrasting strategies in xylem construction. Earlywood Dh ranged from $224.9 \pm 26.5 \mu\text{m}$ (Lednice) to $236.0 \pm 29.5 \mu\text{m}$ (Pohansko), and Kh from $4.51 \times 10^{-7} \pm 1.74 \times 10^{-7}$ (Lanzhot) to $5.31 \times 10^{-7} \pm 2.55 \times 10^{-7}$ (Pohansko). CD was relatively stable at $\sim 6.1\text{--}6.4 \text{ mm}^{-2}$. In latewood, Dh dropped sharply to $\sim 73 \mu\text{m}$, and Kh declined by two orders of magnitude to $\sim 5.3\text{--}5.9 \times 10^{-9}$, while CD increased to $8.9\text{--}15.4 \text{ mm}^{-2}$ (Table. 4), marking a shift toward safer xylem.

Ash also exhibited the highest MLA in earlywood at Pohansko ($36,704 \pm 9,598 \mu\text{m}^2$), with slightly lower values at Lanzhot and Lednice ($\sim 33,700\text{--}33,800 \mu\text{m}^2$). In latewood, MLA dropped to $\sim 2,600 \mu\text{m}^2$ (Table. 4) across sites, indicating a strong seasonal anatomical adjustment.

In Oak, earlywood Dh was larger than in Ash, ranging from $283.9 \pm 55.6 \mu\text{m}$ (Pohansko) to $305.8 \pm 27.6 \mu\text{m}$ (Lanzhot), but Kh was comparable or slightly higher (5.68e^{-7} to 7.36e^{-7}) with greater variability. CD remained low ($4.5\text{--}5.6 \text{ mm}^{-2}$). Latewood Dh narrowed dramatically to $\sim 49\text{--}59 \mu\text{m}$, while CD rose sharply to $\sim 40\text{--}47 \text{ mm}^{-2}$, much higher than in Ash. Kh fell accordingly to $\sim 4.4\text{--}8.0\text{e}^{-9}$ (Table 4). Despite this decline, oak exhibited significantly higher latewood Kh than ash ($p < 0.05$).

Table 4: Growth and anatomical traits for ash and oak trees from three locations (Lanzhot, Lednice, and Pohansko). The table includes tree-ring width (TRW, mm), mean ring width (WIDTH, mm), mean lumen area (MLA, μm^2), hydraulic diameter (Dh, μm), and hydraulic conductivity (Kh, $\text{m}^4\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$). Measurements are provided separately for earlywood (EW) and latewood (LW). Each value is expressed as mean $\pm$ standard deviation.

Type	Species	Location	MRW (mm)	MLA (μm^2)	Dh (μm)	Kh ($\text{m}^4\cdot\text{s}^{-1}\cdot\text{Pa}^{-1}$)
EW	Ash	Lanzhot	1.09 ± 0.40	33492.95 ± 6764.86	227.21 ± 16.43	$(4.40 \pm 1.58) \times 10^{-7}$
		Lednice	1.19 ± 0.27	33496.52 ± 7967.81	224.17 ± 24.96	$(4.91 \pm 1.46) \times 10^{-7}$
		Pohansko	1.08 ± 0.38	36704.37 ± 9598.19	235.99 ± 29.46	$(5.31 \pm 2.55) \times 10^{-7}$
		All	1.11 ± 0.37	34251.21 ± 7951.57	228.45 ± 22.92	$(4.75 \pm 1.86) \times 10^{-7}$
	Oak	Lanzhot	0.82 ± 0.34	57897.40 ± 15829.47	305.32 ± 27.38	$(7.24 \pm 2.94) \times 10^{-7}$
		Lednice	0.58 ± 0.27	53677.87 ± 17436.85	286.71 ± 39.15	$(5.55 \pm 3.46) \times 10^{-7}$
		Pohansko	0.57 ± 0.22	56288.76 ± 18301.86	289.80 ± 31.26	$(4.32 \pm 1.95) \times 10^{-7}$
		All	0.68 ± 0.32	56067.45 ± 17015.13	295.53 ± 33.89	$(6.04 \pm 3.18) \times 10^{-7}$
LW	Ash	Lanzhot	1.19 ± 0.99	2491.08 ± 977.54	71.03 ± 17.02	$(4.48 \pm 3.54) \times 10^{-9}$
		Lednice	1.26 ± 0.64	2561.50 ± 745.76	72.19 ± 14.24	$(5.29 \pm 3.89) \times 10^{-9}$
		Pohansko	0.67 ± 0.53	2549.66 ± 983.20	70.33 ± 17.30	$(4.69 \pm 4.69) \times 10^{-9}$
		All	1.09 ± 0.84	2524.30 ± 919.21	71.19 ± 16.35	$(4.75 \pm 3.93) \times 10^{-9}$
	Oak	Lanzhot	0.82 ± 0.53	1781.52 ± 453.17	56.46 ± 13.97	$(6.44 \pm 4.65) \times 10^{-9}$
		Lednice	0.65 ± 0.54	1481.19 ± 377.27	49.65 ± 10.41	$(3.66 \pm 3.31) \times 10^{-9}$
		Pohansko	0.63 ± 0.41	1461.41 ± 318.69	47.83 ± 8.75	$(4.20 \pm 3.76) \times 10^{-9}$
		All	0.71 ± 0.51	1602.09 ± 426.33	52.06 ± 12.29	$(4.95 \pm 4.22) \times 10^{-9}$

4.1.2.2 Climate and carry-over effects on tree anatomy

MLA was log-transformed prior to analysis due to its skewed distribution, ensuring more robust interpretation of model outputs. Temperature showed a positive, though non-significant, effect on MLA in both Ash ($p = 0.155$) and Oak ($p = 0.200$) (Table. 5; Fig. 18), suggesting a potential trend towards wider vessels under warmer conditions. Precipitation had a significant non-linear effect in Ash ($\text{edf} = 1.98$, $F = 3.00$, $p = 0.045$) (Table. 5; Fig. 18), with MLA increasing under moderate rainfall before declining at higher levels. In Oak, the effect of precipitation was weak and non-significant ($p = 0.227$).

Table 5. Summary of Generalized Additive Model (GAM) results for the effects of environmental and temporal variables on mean lumen area (MLA) and hydraulic diameter (Dh) for ash and oak.

Species	Trait	Term	Estimated Degrees of Freedom	Reference Degrees of Freedom	F-statistic	p.value
Ash	MLA	Precipitation	1.99	2.308	3.061	0.042
		Groundwater	1	1.001	0.29	0.590
		Mean Temperature	1	1	2.074	0.150
		Tree Size	2.592	3.261	3.215	0.018
		Log-transformed Latewood lumen area from previous year	1	1	3.477	0.063
		Site (Random Effect)	11.481	12	27.477	0.000
		Year (Random Effect)	15.425	51	0.467	0.014
Oak	MLA	Precipitation	1.766	2.071	1.155	0.308
		Groundwater	1	1	6.576	0.011
		Mean Temperature	1	1	1.978	0.160
		Tree Size	3.419	4.353	2.191	0.091
		Log-transformed Latewood lumen	2.131	2.733	2.361	0.064

		area from previous year				
		Site (Random Effect)	12.032	13	20.887	0.000
		Year (Random Effect)	15.135	58	0.382	0.021
Ash	Dh	Precipitation	1.723	1.887	2.323	0.161
		Groundwater	1	1.001	0.2	0.655
		Mean Temperature	1	1	4.013	0.046
		Tree Size	2.78	3.482	3.711	0.008
		Log-transformed Latewood hydraulic diameter from previous year	1.001	1.001	1.549	0.214
		Site (Random Effect)	11.651	12	43.949	0.000
		Year (Random Effect)	26.857	51	1.396	0.000
Oak		Precipitation	1.196	1.305	0.885	0.295
		Groundwater	1.048	1.081	8.476	0.003
		Mean Temperature	1.001	1.001	2.084	0.149
		Tree Size	2.936	3.785	1.38	0.220
		Log-transformed Latewood hydraulic diameter from previous year	5.12	6.298	3.698	0.001
		Site (Random Effect)	12.215	13	26.543	0.000
		Year (Random Effect)	18.144	58	0.502	0.006

Groundwater depth was not a significant driver of MLA in ash, though oak showed a slight declining trend ($p = 0.059$), hinting at a possible sensitivity. Tree size significantly

influenced MLA in ash ($\text{edf} = 2.59$, $F = 3.21$, $p = 0.02$) (Table. 5; Fig. 18), where lumen size increased with tree size before levelling off. Oak showed a similar, though statistically marginal, pattern ($p = 0.09$). Previous year's latewood MLA did not significantly affect current-year MLA in ash ($p = 0.07$) or oak ($p = 0.06$). However, the previous year's latewood Dh was a significant predictor in oak ($p < 0.05$) (Table. 5).

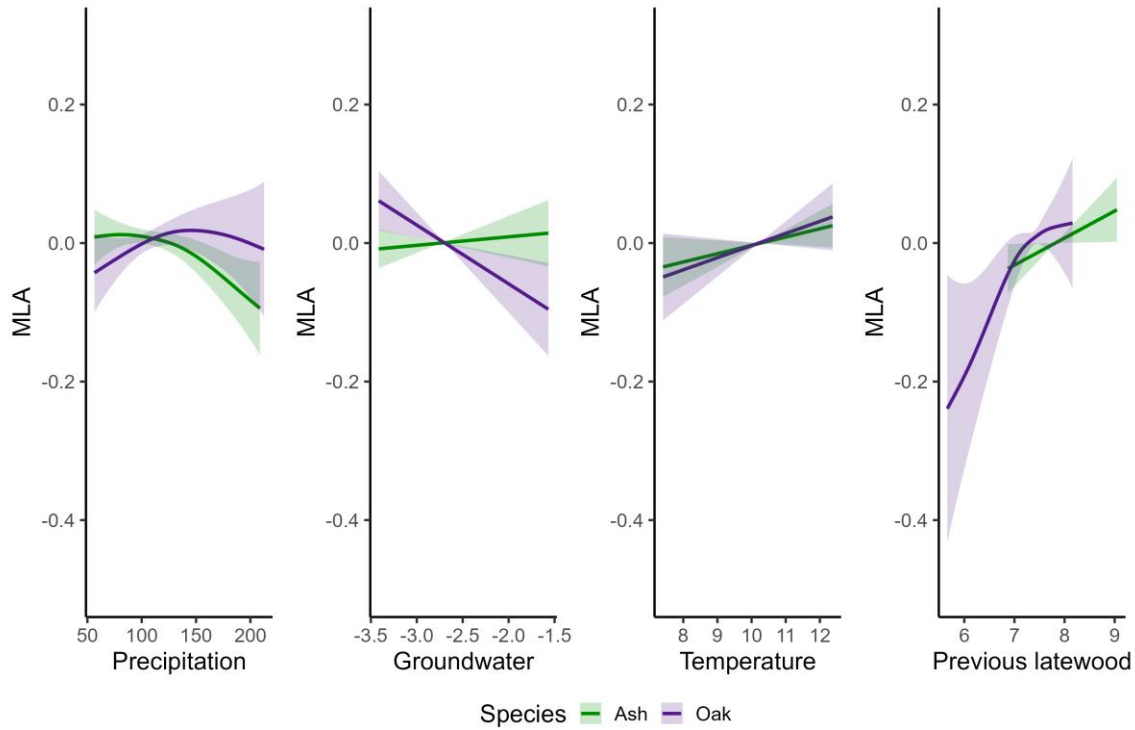


Figure 18. Partial regression plots from Generalized Additive Mixed Models (GAMMs) showing the effects of precipitation, groundwater, temperature, and previous latewood area on the log-transformed mean lumen area (MLA) for ash (green) and oak (purple). Solid lines represent modeled trends, and shaded areas indicate 95% confidence intervals. Note that partial residuals are plotted with free y-axis scales.

Correlation strength between growth and hydraulic traits varied by both species and site. In Ash, the highest correlations were observed at Pohansko, with EWW and MLA ($r = 0.78$), TRW and Kh ($r = 0.60$), and EWW and Dh ($r = 0.52$). At Lanzhot and Lednice (Fig. 19), correlations were generally weaker; the strongest values were EWW and MLA ($r = 0.67$) at Lanzhot and TRW and MLA ($r = 0.05$) at Lednice. Negative correlations were

found between CD and Dh across all sites, most pronounced at Pohansko ($r = -0.59$) (Fig. 19).

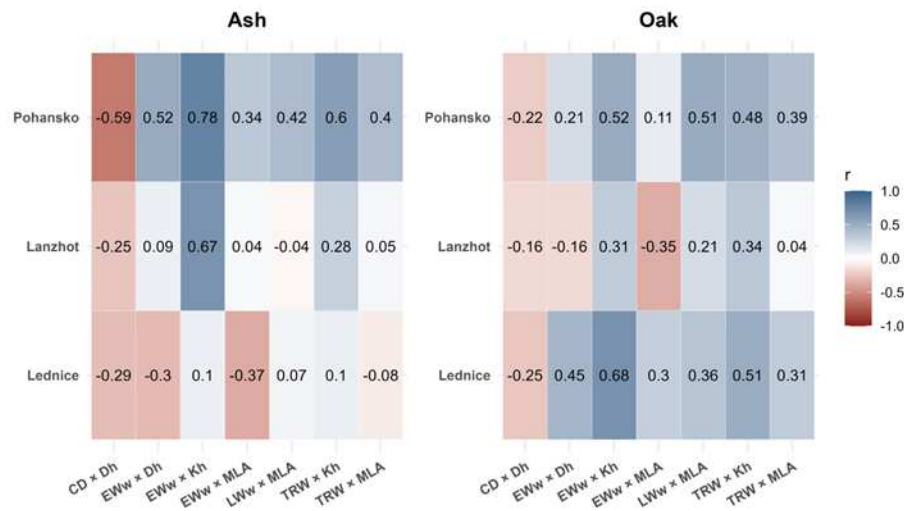


Figure 19. Heatmaps of Pearson correlation coefficients between pairs of xylem anatomical and growth traits for ash (left) and oak (right) across three floodplain forest sites: Pohansko, Lanzhot, and Lednice. Trait combinations include earlywood (EW) and latewood (LW) vessel characteristics such as vessel hydraulic diameter (Dh), hydraulic conductivity (Kh), mean lumen area (MLA), vessel density (CD), tree-ring width (TRW), earlywood width (EWW), and latewood width (LWW). Color intensity reflects the strength and direction of the correlation, with blue indicating positive and red indicating negative relationships.

In Oak, correlations were more uniform across sites. EWW and MLA were positively correlated at all locations, with the strongest association at Lednice ($r = 0.3$) (Fig. 19), followed by Pohansko ($r = 0.11$) and Lanzhot ($r = -0.35$) (Fig. 19). EWW and Kh, TRW and Kh and TRW and MLA also showed consistently positive correlations across sites. CD and Dh were negatively correlated at all sites.

Overall, correlation strength and direction varied more across sites in Ash than in Oak, with Pohansko consistently showing the strongest values in Ash. Oak showed more stable correlations across traits and sites

4.1.3 Effects of artificial flooding

At Pohansko, which has been subject to artificial flooding since 2011, TRW and hydraulic traits were analyzed for two tree species, ash and oak. In ash, there was no significant difference in TRW before and after the onset of flooding. In contrast, oak showed a statistically significant increase in TRW following the introduction of artificial flooding. Analysis of the log-transformed MLA for earlywood and latewood in both species revealed no significant changes between the periods before and after 2011 (Fig. 20). Similar results seen with no significant differences were observed for Dh in both species (Fig. 20).

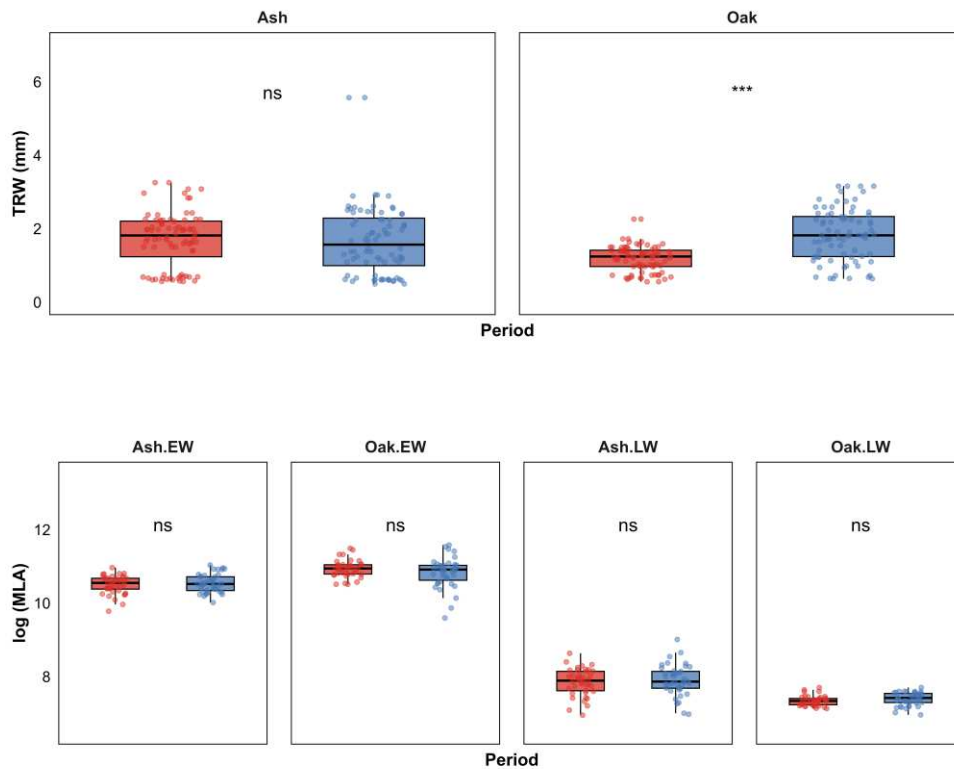


Figure 20. Tree-ring width (TRW) and earlywood (EW) and latewood (LW) mean lumen area (log transformed) before and after drought events for ash and oak. Red and blue indicate pre- and post-drought periods; points represent individual observations. Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

A segmented GAMM of TRW over time indicated that the temporal change of radial growth in Pohansko was not significant before 2011 but became statistically significant after 2011 ($p < 0.01$) (Fig. 26), with an evident increase in TRW values post-flooding.

4.2 Continental-scale patterns in oak

4.2.1 Growth trends across Europe

The chronologies of tree rings in this dataset cover a broad temporal range, with the earliest chronologies dating back to 1709 (Stupnički Lug, Croatia) and the most recent chronologies extending to 2023. This vast temporal coverage allows us for a comprehensive assessment of long-term growth dynamics and climatic sensitivity across diverse European regions. Statistical indicators demonstrate distinct regional variations in chronology quality, with southern sites—especially in Croatia, Hungary, and the Czech Republic, showing the most robust and coherent tree-ring signals. These chronologies exhibit consistently high mean inter-series correlation (R_{bar}), often above 0.6 (Fig. 21), signifying robust synchronicity among individual growth series. Gleichläufigkeit (Glk), indicative of synchronisation in annual growth fluctuations, is similarly heightened in the south, generally exceeding 0.7. Expressed Population Signal (EPS) values are consistently elevated in these areas, frequently exceeding 0.9 (Fig. 21), so affirming the strong representativeness of the chronologies. Autocorrelation (AR) levels in the southern region are typically elevated, indicating increased year-to-year growth persistence. Additionally, autocorrelation (AR) values are typically higher in the south, indicating greater persistence of growth from year to year.

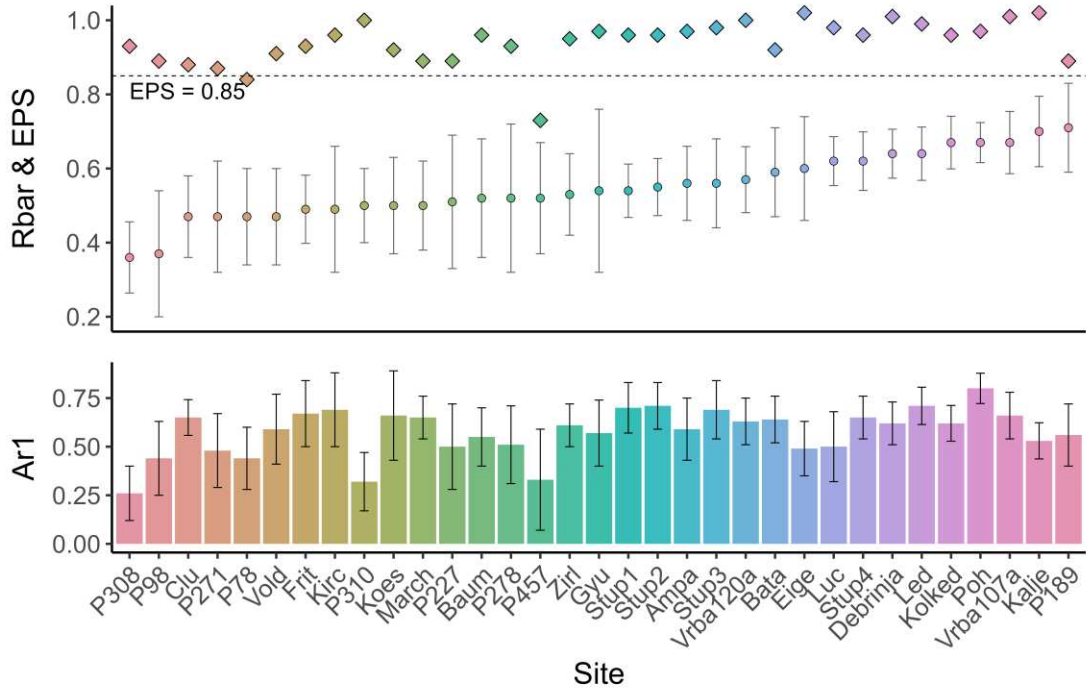


Figure 21. Chronology statistics for all study sites. The top panel shows inter-series correlation (Rbar) as points with error bars and Expressed Population Signal (EPS) as diamonds. The dashed horizontal line indicates the commonly accepted EPS threshold of 0.85. The bottom panel displays first-order autocorrelation (Ar1) as bars with standard error. Sites are ordered by increasing Rbar, and colours distinguish individual sites across the temperature gradient.

On the other hand, sites in northern Poland typically have weaker chronology statistics. Rbar values exhibit lower levels and greater variability, with multiple sites (Jeziorany and Zielonka) registering below 0.5. Glk levels, though satisfactory, generally reside within the moderate range (0.6–0.7) (Fig. 21), and EPS, while predominantly exceeding the crucial barrier of 0.85, frequently nears this lower limit.

High-altitude locations in Austria display intermediate attributes. Some chronologies, such as Eigenhofen, exhibit signal intensity comparable to southern sites with elevated Rbar and EPS values, but others, like Volders, demonstrate diminished coherence and more variability in AR.

1.2.2 Geographical variation in climate–growth sensitivity

Across the 33 populations, climate-growth responses varied considerably (Fig. 22 and Figs. B1–B4). Overall, current season climate, particularly SPEI and P, exerted a stronger influence on radial growth than previous-season variables. In total, 53 significant correlations ($p < 0.05$) were observed between population-level growth and these two climatic drivers (Figs. B1–B2). Regarding the influence of temperature, a notably high number of populations responded significantly to current Tmax (22 out of 33; Fig. B3), whereas current Tmin influenced fewer number of populations (13 out of 33; Fig. B4). In contrast, previous-season climate had a weaker influence on growth, although water availability (i.e., SPEI and precipitation) was significantly correlated with growth in 39 population-variable combinations, while Tmax and Tmin showed 27 such correlations. High populations showed the highest proportion of significant ($p < 0.05$) correlations across climatic variables, followed by southern populations, while northern populations displayed the lowest proportion of significant climate-growth relationships (Fig. 22 and Figs. B1–B4).

Both positive and negative correlations were observed across populations and within each cluster, however significant trends with MAT were identified (Fig. 22). Responses to all previous-season climate variables showed significant trends with MAT, whereas among current-season variables, only Tmax exhibited a significant relationship ($p < 0.05$). For SPEI and P from the previous season, rmax showed a significant negative relationship with MAT, with correlations shifting from positive in colder sites to negative in warmer sites. Current-season SPEI and P were mostly positively related to growth, however, several populations in all clusters showed negative correlations, weakening the overall trend with MAT and resulting in a non-significant slope ($p > 0.05$). Conversely, rmax values for previous-season Tmax and Tmin showed significant positive relationships with MAT; current-season Tmax exhibited the opposite pattern, with correlations shifting from positive at colder sites to negative in warmer regions, while current-season Tmin followed a similar trend, but slope was not statistically significant ($p > 0.05$; Fig. 22).

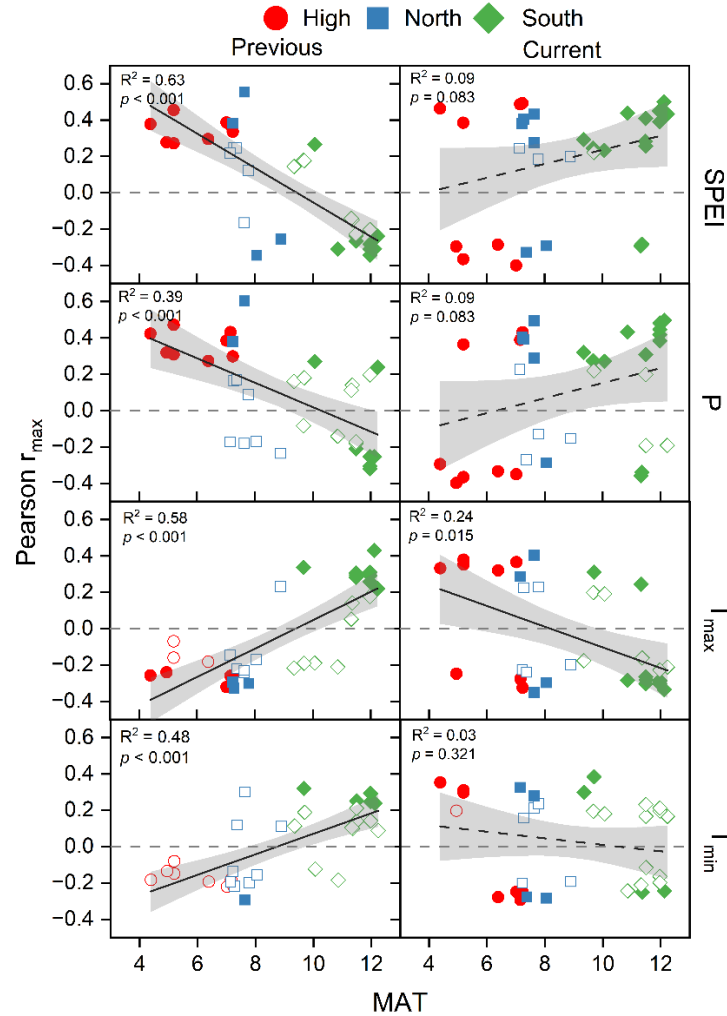


Figure 22. Relationship between the maximum climate–growth correlation (r_{max}) and mean annual temperature (MAT) across study sites. For each site, r_{max} represents the strongest absolute Pearson correlation between tree-ring width indices (TRWi) and seasonal climate variables, including the Standardized Precipitation Evapotranspiration Index (SPEI), precipitation (P), maximum temperature (T_{max}), and minimum temperature (T_{min}), calculated separately for the previous and current growing seasons. Open and closed symbols indicate non-significant and significant correlations at the $p < 0.05$ level, respectively. Regression lines show overall trends across populations, with solid lines representing significant and dashed lines representing non-significant relationships; shaded areas indicate 95% confidence intervals (Stojanović et al., 2025).

Spatial patterns of climate–growth relationships reveal that tree sensitivity to moisture and heat is jointly shaped by elevation and latitude (Fig. 23 and Fig. B5). However, MAT consistently emerged as the most significant predictor across different climate parameters (Table B1). Models of previous

growing - season climate explained 37-66% of the variance, revealing distinct spatial patterns. In contrast, models of water availability (SPEI and P) effects, particularly for the current growing season, exhibited very low explanatory power, with an adjusted R^2 of only 6.4% (Table B1).

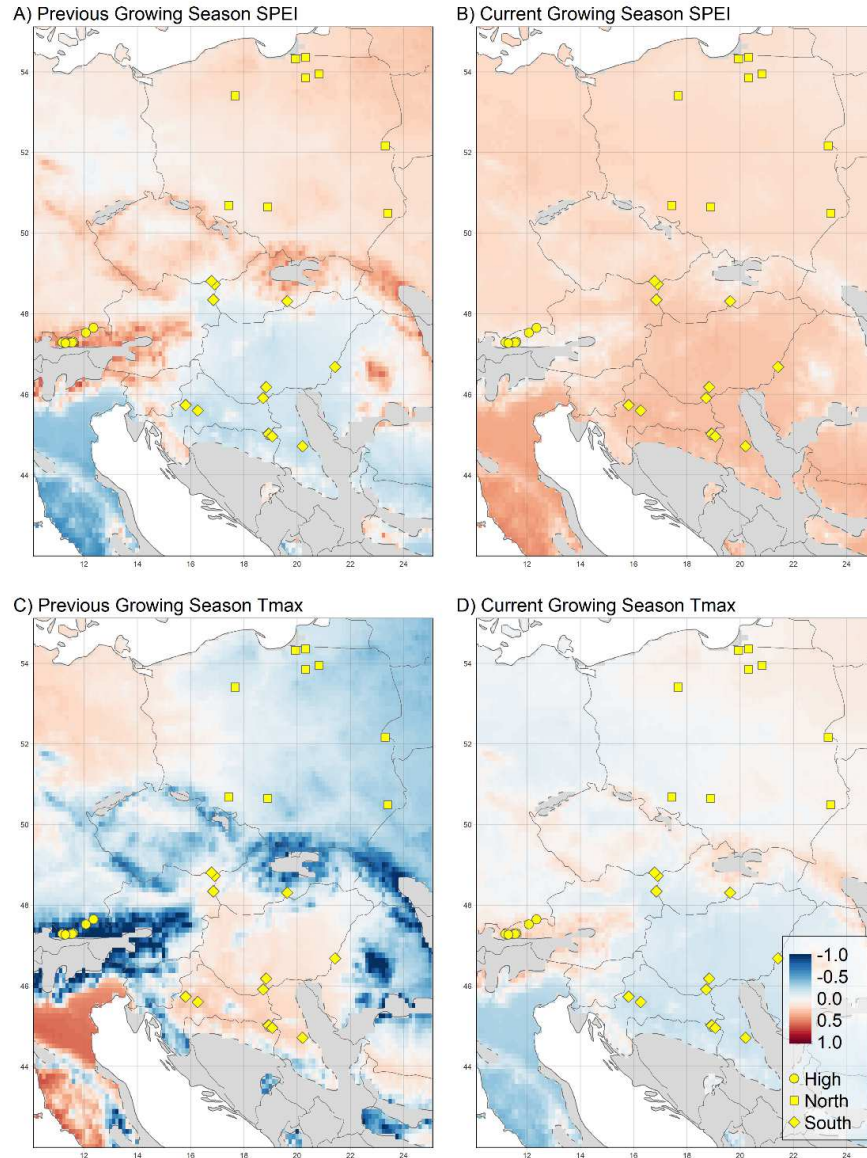


Figure 23. Spatial patterns of climate sensitivity in oak. The upper panels display spatially interpolated correlations between tree growth and the Standardized Precipitation Evapotranspiration Index (SPEI) for the (a) previous and (b) current growing seasons, while the lower panels show the influence of (c) previous and (d) current season maximum temperature (Tmax) on growth (Stojanović et al., 2025).

These patterns thus reflect the influence of both previous- and current-season climatic conditions on current-year growth. Overall, upper-elevation populations (i.e., high) showed stronger sensitivity to current-season temperature, and weaker sensitivity to current-season SPEI and precipitation. For SPEI and P, a highly positive effect was observed from the previous season, but this effect diminished or reversed in the current season. Tmax exhibited a contrasting pattern, with a generally negative effect from the previous season and a positive effect in the current season. Tmin showed a weaker negative effect in the previous season, shifting to a markedly stronger negative effect in the current season.

In contrast, patterns in southern and northern populations differed from those at high elevation and were more strongly influenced by moisture than temperature. Generally, the strength of climate-growth correlations declined from south to north (Fig. 23 and Fig. A5). Current-season SPEI and precipitation had a positive effect on growth across the gradient, but the strength of these relationships weakened toward the north. Notably, the effect of previous-season SPEI and precipitation shifted along the latitudinal gradient, changing from negative in the south to positive in the north. Overall, temperature-growth correlations were relatively weaker. Current-season Tmax and Tmin showed weak negative effects in southern populations and weak positive effects in northern populations. Notably, previous-season Tmax and Tmin showed opposite effects compared to the current season (Fig. 23 and Fig. B5).

4.2.3 Resilience to drought across the species range

Negative pointer year analysis showed that southern populations experienced the highest number of extreme growth reductions (Fig. 11). To assess resilience, we calculated resistance, recovery, and overall resilience by pooling all disturbance events that led to significant radial growth declines across the three population groups (Fig. 24). All groups showed generally low resistance, but significant differences were found between all group pairs ($p < 0.05$). High-elevation populations had the highest resistance (mean = 0.71), followed by southern (0.67) and northern populations (0.59).

In contrast, recovery was lowest in the high-elevation group (mean = 1.37), significantly lower than in southern (1.47) and northern (1.83) populations, which did not differ from each other ($p < 0.05$). Despite these variations, all groups' overall resilience was comparable, with mean values near 1 and no significant variances ($p > 0.05$) (Fig. 24).

Comparing the three groups' responses to disturbance to a theoretical standard of full resilience, the fitted curves illustrating the link between recovery and resistance provided further information (Fig. 25). The negative power function most accurately represented the observed patterns. With the lowest residual error and the highest correlation coefficient, populations at higher elevations demonstrated the best match. Whereas southern populations were in the middle, northern populations had the poorest and most inconsistent fit. These patterns were also reflected in NSE values, where only the high-elevation group showed a comparatively best agreement by positive NSE, indicating good alignment with the theoretical recovery–resistance curve. Southern and northern groups had lower NSE values, suggesting greater deviation from the theoretical resilience model. The point where each group crossed the line of full theoretical resilience differed: high-elevation populations at a resistance value of 0.8, southern at 0.77, and northern at 0.51 (Fig. 25).

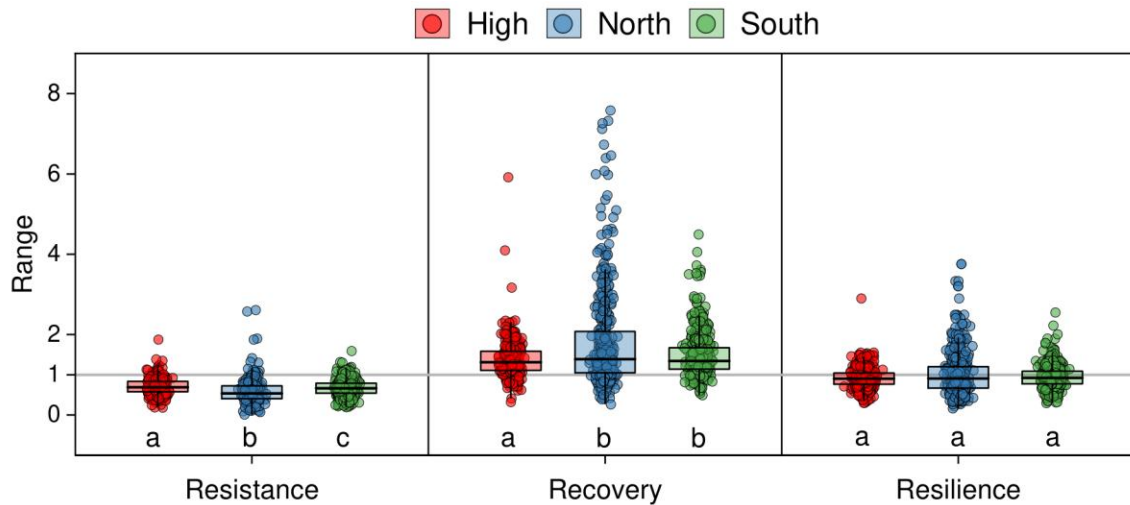


Figure 24. Resilience indices of pedunculate oak across three biogeographic clusters: High (red), North (blue), and South (green), calculated from all disturbance events that caused extreme reductions in radial growth. The plot displays distributions of recovery, resilience, and resistance values. Boxplots show the median, interquartile range, and whiskers extending to 1.5 times the interquartile range, with individual points representing tree-level data. Different letters indicate statistically significant differences between clusters according to the Mann–Whitney U test ($p < 0.05$) (Stojanović et al., 2025).

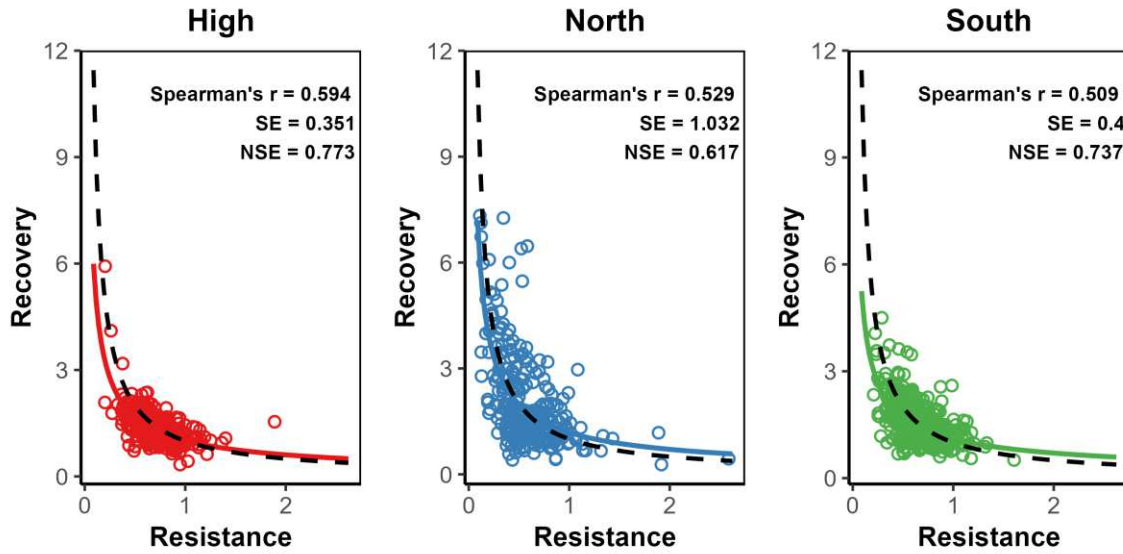


Figure 25. Comparison of the observed relationship between resistance and recovery across three biogeographic clusters: High (red), North (blue), and South (green), based on all disturbance events that caused extreme radial growth reductions in oak. The black dotted line represents a theoretical model of full resilience (Schwarz et al. 2020), and vertical lines indicate the resistance values where each empirical curve intersects this reference model (Stojanović et al., 2025).

5. Discussion

5.1 Growth sensitivity and resilience to drought and hydrology

5.1.1 Climate and groundwater influence on tree ring width

Our findings showed that radial tree growth and climate had species-specific associations, indicating different ecophysiological responses to environmental change. We consistently found greater associations between radial development and water-related indicators such as precipitation, groundwater, and SPEI than with temperature. The responses of the two species were notably different. Predominant significant correlations emerged for ash, but oak exhibited only a limited number of meaningful associations.

The correlation between temperature and radial growth was generally modest across sites. However, elevated temperatures can increase PET, leading to enhanced surface water loss and greater drought stress. The water deficit when PET surpasses precipitation can lower soil moisture and worsen dry situations. Spring and summer in our study area are often characterised by a negative water balance (precipitation – PET) (Fischer et al., 2023; Kowalska et al., 2020), which is also reflected in our climatic data (Figs. 6, 7). Furthermore, heightened evapotranspiration may exacerbate the inverse relationship noted between temperature and groundwater levels (Fig. S6; Fischer et al., 2023), a tendency anticipated to grow in the future due to the ongoing increase in temperatures (Fig. 6). Conversely, summer precipitation can enhance groundwater recharge, hence potentially augmenting water availability for forests (Haidu and Nistor, 2020). This effect may become significant due to the little rise in summer precipitation noted in our region (Fig. 6). In summary, although climate affects seasonal groundwater dynamics, we contend that human activities are probably the primary cause of groundwater depletion at our study location (Klimo et al., 2013; Šenfeldr et al., 2021).

Ash exhibited a robust positive association between radial growth and precipitation during April and May (Fig. 13), consistent with other research (Mikac et al., 2018; Šenfeldr et al., 2021). Ample water availability during the commencement of cambial activity and earlywood formation enhances photosynthetic rates, hence facilitating increased radial

development (Patrick et al., 2009). Groundwater considerably affected ash growth from May to the end of the growing season (Fig. 14). This constant water source apparently promotes hormonal activity and cellular differentiation, contributing to the formation of wider rings (Kozłowski and Pallardy, 1997). Under moist conditions, ash demonstrates increased stomatal conductance, enabling extended photosynthesis and continuous secondary growth. In contrast, oak demonstrates a reduced dependence on precipitation, as indicated by its weak and largely insignificant correlations (Fig. 13), likely due to its broad root system and relatively anisohydric regulating approach.

The SPEI-3 index exhibited the most robust correlation with radial growth (Fig. 14). The SPEI composite drought index accounts for complicated climate-water interactions by integrating temperature, precipitation, and PET (Beguería et al., 2014). The divergent drought responses of oak and ash are likely attributable to their distinct roots patterns and water utilisation methods. Oak, characterised by its extensive root system and ectomycorrhizal relationships (Allen, 2015), generally absorbs water from deeper soil strata, although may adjust its intake in response to surface moisture levels (Kreuzwieser and Gessler, 2010; Tatarinov et al., 2008). As a result of this adaptive flexibility, oak is able to maintain its stomata open even during dry periods (Szatniewska et al., 2022). This may be the reason why the relationships with groundwater, precipitation, and drought indices are less significant (Figs. 13,14).

Due to its reliance on superficial soil moisture and its adaption to floodplain environments, ash is more vulnerable to fluctuations in surface water availability (Sánchez-Pérez et al., 2008). Its isohydric method, which involves closing stomata during water stress to maintain a constant leaf water potential (Pallardy, 2008; Szatniewska et al., 2022), reduces xylem cavitation while also limiting carbon absorption. As a result, ash has stronger ties to precipitation, groundwater, and SPEI, indicating increased sensitivity to water availability. In places where groundwater loss was more severe, these correlations were more apparent, particularly for ash. At the Lednice site, however, where groundwater levels were rather constant, correlation values were negligible (Fig. 7).

5.1.2 Effect of groundwater regime on the temporal development of climate-growth relationships

Significant changes in tree sensitivity were revealed during the study period by the typically inconsistent associations we found between radial growth and hydroclimatic factors (Figs. 15,16). Interestingly, there was a high correlation between radial growth and short-term water availability indices, especially SPEI-3. Previous research on ash (Heklau et al., 2019; Koprowski et al., 2018; Roibu et al., 2020; Šenfeldr et al., 2021) and oak (Árvai et al., 2018; Čufar et al., 2014; Filippo et al., 2010; Mikac et al., 2018; Stojanović et al., 2015) have shown similar trends.

Both species showed growing relationships between TRW and SPEI-3 and groundwater levels in our dataset at groundwater decline sites, namely in Lanzhot and Soutok (Fig. 16). Oak, on the other hand, showed only weaker associations for a smaller period of the growing season. From April to September, the sensitivity of tree ring to hydroclimatic parameters increased, mainly in these drier sites. This alligns with previous findings that floodplain forests in these places are more vulnerable to hydroclimatic change (Šenfeldr et al., 2021). On the other hand, we observed a declining trend in climate sensitivity at the Lednice site, suggesting that more stable local groundwater conditions could mitigate the adverse effects of rising temperatures (Skiadaresis et al., 2019).

Precipitation in May was the most notable precipitation variable, exhibiting the highest and most sustained upward association with radial expansion across time (Fig. 15). The implication that water availability is especially crucial for radial growth in ring-porous species during the early phases of cambial activity and leaf expansion is supported by this (Noyer et al., 2023; Pérez-de-Lis et al., 2018). Elevated spring rainfall reduces VPD, which enhances stomatal conductance, photosynthesis and secondary growth, an effect especially evident in May.

During summer and autumn, ash at all sites maintained strong and stable correlations with groundwater levels throughout the entire period of observation. This relationship became more pronounced following the onset of river modifications in 1968 (Fig. 16). Oak, by contrast, showed no significant correlation with groundwater until 1988, the year river

modifications were completed. From that year onwards, a significant positive relationship emerged. This delayed response likely reflects oak's higher tolerance to water stress, unlike the greater vulnerability of ash to water scarcity.

Interestingly, the ring width – groundwater correlation intensified over time at the Soutok, Lanzhot, and Pohansko sites, whereas a weakening trend was observed at Lednice (Fig. 16). Trees at this site likely had year-round access to groundwater due to their well-established, deeper root systems, making them less susceptible to fluctuations in groundwater availability.

It is important to note that our analysis did not account for stand-specific factors such as inter-tree competition, mast years, or insect outbreaks, all of which can substantially influence growth-climate relationships. For instance, Hacket-Pain et al. (2015) reported that mast events could result in exceptionally narrow rings, while Scharnweber et al. (2011) linked insect outbreaks to marked growth suppression. Incorporating such biotic and stand-level variables into future research could provide a more comprehensive understanding of the drivers behind tree growth variability.

Overall, the trees in our study relied primarily on precipitation during the early part of the growing season (Figs. 13, 15). However, from late spring through to the end of the growing season, a distinct shift occurred. During this time, groundwater became the dominant water source influencing radial growth (Figs. 14, 16).

5.1.3 Impact of drought on tree resilience components

Our results demonstrate that both oak and ash growth are adversely affected by severe droughts during the growing season, however the degree of the impact differs between plots and species (Fig. 17). Both species showed decreased growth stability in locations with diminishing groundwater levels, as indicated by lower values for all resilience components. The findings of Lévesque et al. (2016), who found that reduced groundwater availability can impede tree development, are in line with our findings. The diminished resistance and resilience observed at these locations imply that groundwater availability, which is still an essential resource for maintaining physiological processes during dry

spells, greatly influences how trees react to drought (Oosterbaan and Nabuurs, 1991; Skiadaresis et al., 2021).

In 1990 and 2012, ash displayed decreased resistance values in every plot (Fig. 17), highlighting its susceptibility to drought stress. This is consistent with previous research that has shown how sensitive the species is to water scarcity (Mikac et al., 2018; Pericolo et al., 2023; Šenfeldr et al., 2021). These findings support our third hypothesis, which states that oak is more resilient and typically less vulnerable to the negative consequences of drought. Most notably, lower groundwater levels and less spring precipitation combined may have contributed to poorer resilience values in 2003 and 2012 (Fig. 6, 7), which points to oak's particular sensitivity to spring drought.

At Pohansko, we observed improved resilience and recovery for both species during the 2012 drought. The seasonal water shortage was lessened by a brief artificial water recharge event, which seems to be connected to this. By temporarily raising groundwater levels, artificial flooding can mitigate drought stress in floodplain forests, as shown by Klimo et al. (2013).

The resilience of individual trees varied significantly despite this overall pattern. This heterogeneity most likely results from fluctuating growing circumstances in mixed stands, where variations in soil structure, groundwater depth, and microclimate can be noticeable, particularly in larger plots. In certain regions, extended waterlogging can also be harmful, possibly resulting in a reduction in development and oxygen availability (Kreuzwieser et al., 2004).

5.2 Hydraulic strategy and xylem plasticity in response to water deficit and flooding

5.2.1 Species-Specific Tree Responses to Water, Temperature, and Legacy Effects

Temperature had relatively minor and statistically insignificant effects on earlywood vessel characteristics in both ash and oak (Fig. 18). Nevertheless, its ecological role is evident in regulating the onset of cambial activity, which determines the timing and duration of vessel

expansion. In ring-porous species such as oak, earlywood vessel size is primarily governed by the length of the enlargement phase prior to secondary wall lignification, a period constrained by the timing of cambial reactivation (Pérez-de-Lis et al., 2016). As this reactivation is driven by temperature and photoperiod, climatic conditions during late winter and early spring become critical (Tumajer and Treml, 2016). Notably, vessel enlargement in oak commences approximately one month before bud swelling (Pérez-de-Lis et al., 2016; Sass-Klaassen et al., 2011), at a time when the crown is not yet producing significant quantities of assimilates or phytohormones. Consequently, earlywood development relies on carbohydrate reserves accumulated in the previous year (Michelot et al., 2012) and locally synthesised phytohormones (Aloni, 2015), allowing signals from the preceding growing season to be retained in vessel lumen area. Thus, temperature indirectly influences wood formation by setting the growth period during which internal reserves govern early development.

Moisture availability was the primary factor shaping earlywood anatomy, with species-specific responses based on differing water acquisition strategies. In ash, vessel size followed a nonlinear relationship with precipitation (Fig. 18). It increased with moderate rainfall but decreased during prolonged wetter periods. This species depends on surface water early in the growing season. While moderate precipitation promotes larger vessels (Camarero et al., 2021), saturated soils and elevated groundwater levels result in high soil moisture, leading to the formation of numerous smaller vessels and a reduction in total vessel area, which is a common adaptation in floodplain ecosystem (Tardif et al., 2021). Although the correlation between groundwater and vessel characteristics was weak, it remained positive, aligning with the positive relationship between river flow and soil moisture reported by (Camarero et al., 2021). This is likely because elevated groundwater, along with precipitation, enhances soil moisture retention, making water more available to Ash's shallow root system (Sánchez-Pérez et al., 2008). During dry periods, Ash conserves water by closing its leaf pores, a strategy that maintains internal water pressure and supports hydraulic safety (Pallardy, 2008; Tyree and Zimmermann, 2002).

With their deep root systems and associations with ectomycorrhizal fungi (Allen, 2015), Oaks generally prefer water from surface soil layers where enzymatic activity and nutrient

availability are high (Kreuzwieser and Gessler, 2010). However, oak can shift water uptake to deeper layers during drought periods, adapting to changing environmental conditions (Allen, 2015; Tatarinov et al., 2008). This flexibility in water sourcing enables oak to maintain physiological function under fluctuating moisture regimes. In contrast, under floodplain conditions with persistently high groundwater levels, excessive soil moisture may lead to waterlogging. This impairs root respiration, a process essential for energy production and active nutrient uptake. These impairments in root functioning restrict water and nutrient acquisition, thereby constraining photosynthesis and overall growth (Ferner et al., 2012; Kniesel et al., 2015; Kreuzwieser et al., 2004). Such physiological stress may explain the observed reduction in earlywood vessel size associated with rising groundwater levels, as root zone saturation limits the development of large vessels, a response similarly observed by Tumajer and Treml, (2016).

Interannual effects in oak were modest yet discernible, indicating a degree of physiological continuity between growing seasons. Notably, the Dh of latewood vessels formed in the preceding year exerted a significant negative effect on earlywood Dh (Table. 5) in the subsequent year, suggesting a compensatory mechanism, potentially linked to internal resource allocation or the timing of dormancy onset and release (Kitin et al., 2004). A comparable, though statistically non-significant, pattern was observed in *Fraxinus excelsior*. In ring-porous hardwoods, earlywood vessels are typically produced at the onset of the growing season and facilitate rapid sap transport during periods of intense vegetative expansion and elevated humidity (Kitin et al., 2001). These vessels are progressively occluded by tyloses or gums as the season advances (Saitoh et al., 1993), and are generally devoid of water by the end of the growing period (Utsumi et al., 1999). By contrast, latewood vessels, although narrower, sustain slower but more consistent water transport and often remain functional into the following spring (Cochard and Tyree, 1990; Ellmore and Ewers, 1986). The observed inverse relationship between latewood and earlywood Dh suggests a functional trade-off, whereby residual hydraulic capacity from latewood reduces the need for wide earlywood vessels in the subsequent season. This interpretation is further supported by our finding that Oak latewood vessels exhibit significantly higher hydraulic conductivity than those of Ash, indicating greater carry-over hydraulic potential between growing seasons.

Differences in microhabitats also influenced functional traits and growth, likely due to subtle topographic and moisture variation (Peipoch et al., 2023; Schwartz et al., 2020). This heterogeneity may also contribute to anatomical differences, although this was not directly assessed in the present study. In addition to environmental heterogeneity, variation in earlywood and latewood characteristics may also reflect individual tree responses to local competition and crown dynamics. Tree-level factors such as crown size, which is influenced by stand density and competitive pressure, have been shown to affect the ratio of earlywood to latewood, with larger crowns under ideal growth conditions leading to lower proportions of latewood (Domec and Gartner, 2002; Larson, 1969). This underscores the importance of recognizing individual-level variation when interpreting wood formation in relation to site conditions.

Patterns of trait interaction further support these interpretations. In Ash, the relationship between average vessel area and tree ring width varied by location (Fig. 19), likely reflecting its dependence on surface water and the influence of site-specific water status driven primarily by differences in groundwater levels, as precipitation was relatively uniform across sites. Similar observations in radial growth were observed by (Basu et al., 2024; Šenfeldr et al., 2021). Ash's shallow rooting pattern makes it particularly sensitive to variations in surface moisture availability, which is influenced by site-specific differences in groundwater levels.

Oak, by contrast, showed consistent relationships between average vessel area, tree ring width, and earlywood width across all sites (Fig. 19). This consistency reflects a more conservative growth pattern and greater access to deeper water. However, ongoing depletion of groundwater in some areas may challenge this stability. Although no structural decline has yet been observed, prolonged dry conditions combined with reduced groundwater may exceed Oak's compensatory ability, increasing its risk of internal air blockage during drought.

5.2.2 Effect of artificial flood in tree ring and anatomy

Artificial spring flooding resulted in a marked increase in radial growth in oak, particularly at Pohansko, while ash has no significant effect (Fig. 20). These growth differences

occurred without significant changes in xylem anatomy, such as MLA, which remained stable across species and sites.

Although artificial floods did not produce a sustained rise in the groundwater table, they likely enhanced topsoil moisture in the short term. This is supported by findings from Morales et al., (2021), who reported a 47% increase in soil moisture within the upper 10 cm during the first three days of flooding, with minimal further increase thereafter. Their isotopic tracer analysis also showed that moisture in deeper horizons, though initially less influenced, gradually shifted towards the isotopic signature of floodwater over time (Morales et al., 2021). Similarly, Bártová et al., (2021) observed a measurable but short-lived rise in groundwater levels, typically centimetres to decimetres during artificial floods at this site. Although direct soil moisture data were not collected in this study, these findings indicate that artificial floods predominantly affect surface layers.

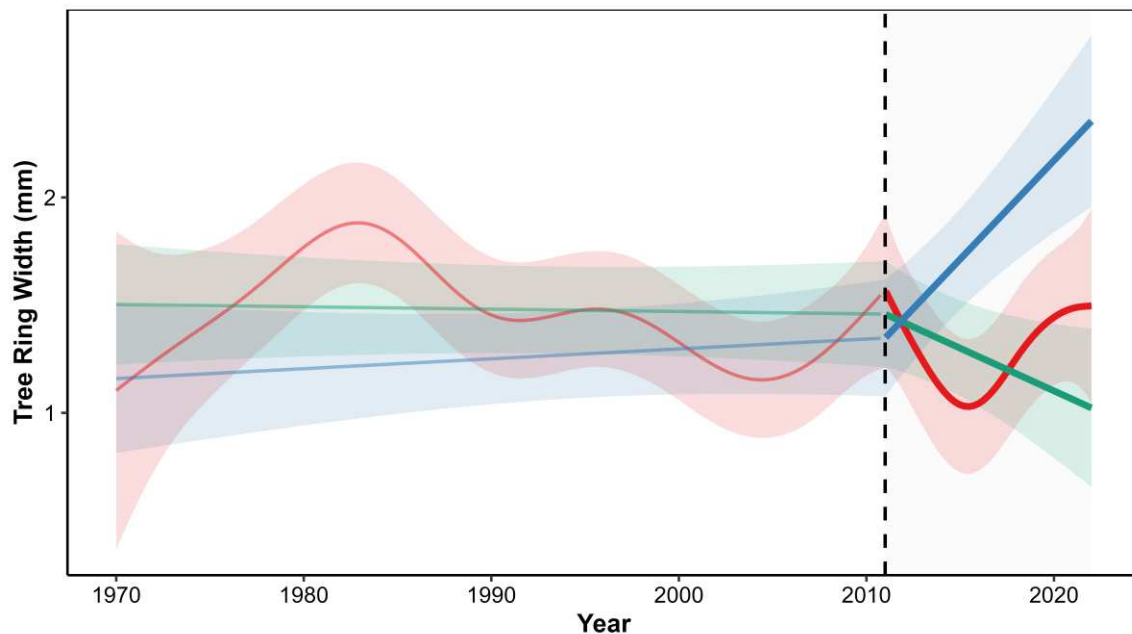


Figure 26. GAMM fits of tree ring width (mm) over time (1970–2022) for three sites: Lanzhot (red), Pohansko (blue), and Lednice (green). A vertical dashed line marks introduction of artificial flood at Pohansko in 2011.

The timing of artificial floods is also a crucial factor. These events are typically conducted in March for a duration of 3 to 7 days, aligning with the onset of cambial activity in oak,

while ash remains dormant. This mismatch with ash's late phenology (Nezval et al., 2020), combined with its reliance on surface water, restricts its ability to benefit from short-lived moisture inputs. In contrast, oak, with its deeper root system (Allen, 2015) and earlier seasonal activity (Nezval et al., 2020), is better positioned to exploit residual subsoil moisture, particularly under the increasingly frequent spring droughts of recent years. Combining these factors helps Oak in Pohansko to increase the radial growth in comparison with other sites (Fig. 26).

The absence of anatomical change in both species suggests the presence of a well-developed, potentially overbuilt conductive system. Mature trees can reportedly lose a substantial proportion of their conductive area without impairing canopy water supply (Dietrich et al., 2018) indicating that growth responses are primarily governed by water availability rather than xylem structure.

Oak, therefore, responded positively to artificial flooding through enhanced radial growth without requiring anatomical adjustment. Ash, by contrast, derived limited benefit, constrained by its physiological traits and the timing of its growth initiation.

5.3 Spatial variation in resilience and climate sensitivity across the temperature gradient

5.3.1 Climate sensitivity across biogeographical regions

Our findings highlight a clear spatial pattern in climate-growth sensitivity among oak populations distributed along wide latitudinal and elevational gradients (Figs. 22 and B1-B4). Growth responses were primarily shaped by current-season climatic conditions, especially the SPEI and precipitation. This highlights the pivotal role of water availability in influencing oak productivity across different biogeographic zones. These results align with previous work emphasising the critical influence of growing-season moisture on radial growth in ash and related temperate oak species.

There were clear changes in climate sensitivity based on latitude and elevation. Populations at higher elevations showed the most constant and clear links between climate and growth. This is probably because the climate is more limited and the ecological ranges are smaller

near the upper limits of elevation in Austria (Garamszegi et al., 2025). On the other hand, population in the north was much less sensitive. Cooler, mesic environments in these areas likely buffer oaks from drought stress, consistent with patterns seen at more humid, buffered sites (Bose et al., 2021; Čufar et al., 2014; Matisons et al., 2012).

Across all population clusters, SPEI and precipitation during the current growing season had a broadly positive impact on growth (Fig. 22). However, this effect was most pronounced in southern populations and tapered off towards northern and high-elevation areas. Interestingly, total precipitation did not vary substantially between southern and northern regions, yet southern sites experienced considerably higher temperatures during the growing season (Fig. 8). This supports the idea that elevated temperatures in the south increase evapotranspiration, exacerbating drought stress during dry spells (Basu et al., 2024; Mikac et al., 2018).

In cooler, wetter regions, particularly at high elevations and, to a lesser extent, in the north precipitation usually exceeds potential evapotranspiration (PET). Here, temperature can exert a stronger influence on growth than water availability (Matisons et al., 2012), although this pattern was highly site-specific (Fig. 22). Overall, temperature showed a more variable and complex effect than moisture availability, exerting a weaker yet regionally distinct influence (Fig. 22). In these colder, moisture-rich zones, warming; particularly increased maximum temperatures may lengthen the growing season and enhance productivity. Conversely, in drier and warmer regions, rising temperatures are more likely to intensify water stress and limit growth (Allen et al., 2010). While higher minimum temperatures in northern areas could encourage earlier onset of growth, this shift may heighten the risk of late frost damage, especially in high-elevation sites (Vitasse et al., 2019).

Climatic conditions during the previous autumn played a relatively minor role (Fig. 23), with only a small number of sites showing significant responses. This supports earlier findings that suggest limited or inconsistent impacts of prior-season climate on oak growth (Móricz et al., 2021; Vanhellefont et al., 2019). As a ring-porous species, oak relies on non-structural carbohydrate (NSC) reserves accumulated the year before, used mainly for earlywood vessel formation initiated before budburst (Michelot et al., 2012). Due to

freeze–thaw damage to the previous year’s vessels, new ones must form swiftly in spring to re-establish sap flow (Noyer et al., 2023; Szatniewska et al., 2022). Hence, favourable late summer to early autumn conditions is essential for building NSC reserves. Interestingly, SPEI and precipitation from the previous season had a negative impact on growth in southern populations (Figs. 22,23), likely due to shallow groundwater tables in these areas (Basu et al., 2024; Mikac et al., 2018). Excessive wetness during this period may reduce growth. During dry late summers, however, photosynthesis may persist even as radial growth declines, supporting the sink limitation hypothesis—where cambial activity is more sensitive than carbon assimilation to drought or heat (Körner, 2015). This allows NSC to accumulate when wood formation slows, potentially boosting growth the following year (Michelot et al., 2012).

The direction and strength of climate-growth relationships varied notably among populations, reflecting substantial heterogeneity in climate sensitivity across the network (Bose et al., 2021). Notably, models incorporating previous-year maximum temperature explained up to 66% of growth variability. In contrast, current-year moisture-based models (SPEI and P) explained less variance, implying a relatively uniform moisture response and greater spatial variability in energy-related constraints (Garamszegi et al., 2025).

These population-level differences likely arise from factors not directly accounted for in this study. Local soil characteristics and their water retention capacity (Kostić et al., 2021), along with differences in groundwater availability (Basu et al., 2024), may significantly influence site-specific growth responses. Additionally, forest management practices, including stand density and competition and tree origin (seed vs coppice) affect allocation patterns such as root-to-shoot ratios, thereby modulating climate-growth responses (D’Orangeville et al., 2025; Stojanović et al., 2017).

5.3.2 Resilience mechanisms and disturbance responses

Our analysis of disturbance responses across oak populations revealed clear variation in resilience dynamics among the three groups (Fig. 24). The degree of growth decline reflected the climate sensitivity of each group. High-elevation populations were the most resistant to drought, followed by those in the south, while northern populations were the least resistant. Recovery, in contrast, followed the reverse pattern. Northern populations

showed the highest ability to recover after disturbance. High-elevation populations recovered the least, and southern ones fell in between. This kind of trade-off between resistance and recovery is common in trees, where stronger initial resistance often comes with slower recovery afterwards (Bose et al., 2021; Dendoncker et al., 2025; Móricač et al., 2021). It likely reflects physiological differences or site-related buffering mechanisms (Schwarz et al., 2020).

Oak is typically described as anisohydric, continuing to transpire during drought by keeping its stomata open (Stojanović et al., 2024; Szatniewska et al., 2022). However, other studies describe an isohydric response, where stomata close quickly once water becomes limited (Schnabel et al., 2022). This variation suggests a strong role of local adaptation. Oak has evolved under a wide range of environmental conditions, which may have led to different stomatal regulation strategies. Northern populations are likely to follow more conservative water-use strategies, which reduce carbon uptake and slow growth, but help preserve water and internal reserves. This may support quicker recovery after disturbance. By contrast, high-elevation and southern populations may rely less on conservation. These groups maintain higher growth during drought by keeping stomata open, yet this can lead to greater stress and slower recovery due to depleted resources (Móricač et al., 2021). Still, recovery in the southern group was not significantly different from the northern one, even though values were lower and less variable (Fig. 24).

Resilience, which includes both resistance and recovery, did not show significant differences among the groups. Mean values stayed close to one across all populations (Fig. 24). This suggests that resilience may be achieved in different ways. Some populations maintain growth during stress, while others recover more quickly once the disturbance passes.

The full resilience framework (Dendoncker et al., 2025; Móricač et al., 2021; Schwarz et al., 2020) offered further insight. High-elevation populations followed the expected resistance–recovery pattern most consistently and had the highest NSE values (Fig. 25). Northern and southern populations deviated more from the theoretical curve, especially where resistance was low. The curves fitted for each group crossed the theoretical line at different points. Northern and southern populations crossed at lower resistance values than

those at high elevation, which indicates stronger recovery even after major declines. This suggests that high-elevation populations follow a more predictable resilience path, while northern populations, despite having good recovery capacity, may show more variation and potential vulnerability (Dendoncker et al., 2025; Schwarz et al., 2020).

6. Integrating water management and climate adaptation in forest management

6.1 Local implications: Managing south Moravian floodplain forests

The findings of this study underscore the pivotal role of hydroclimatic conditions, especially groundwater availability and precipitation in governing the species-specific growth dynamics and resilience of floodplain forests (Skiadaresis et al., 2019). These ecosystems are highly sensitive to water regime alterations, with prolonged groundwater decline posing significant threats to both individual tree vitality and forest structure (Cunningham et al., 2011).

Our findings clearly indicate that drought stress is gradually affecting floodplain forests, with oak exhibiting greater resilience, whereas ash is more vulnerable to diminishing groundwater levels. Artificial flooding demonstrated efficacy (Bártová et al., 2021), though its advantages were species-specific, substantially benefiting oak while providing minimal benefits for ash. This is likely due to oak's deep-rooting system and its ability to access recharged groundwater more effectively (Allen, 2015). In contrast, ash, with its shallower roots, remains dependent on surface moisture that is slightly improved by such interventions (Sánchez-Pérez et al., 2008). Future forest management strategies must emphasize sustainable water regulating practices. The establishment of water retention reservoirs that provide a slow and regulated flow of water can replicate natural hydrological cycles, maintain groundwater levels, and promote the long-term stability of floodplain forest ecosystems. Furthermore, additional research into root morphology and belowground adaptations is crucial for a deeper understanding of species-specific water

acquisition strategies and their role in drought resilience (Čermák and Prax, 2001; Huntley, 2023).

Results from the anatomical and tree-ring analyses reinforce this vulnerability. Oak with its deeper rooting system and dual water-foraging capacity, exhibits greater resilience to declining water tables, maintaining hydraulic conductivity and growth under increasingly arid conditions (Šenfeldr et al., 2021; Stojanović et al., 2015). In contrast, ash shows stronger growth suppression and hydraulic limitations under water stress, making it particularly susceptible to groundwater decline. These contrasting ecophysiological responses underscore the importance of carefully considering the timing of artificial flooding, which need not be uniformly aligned with the water requirements of both species, given their markedly different drought tolerance and adaptive strategies.

Isolated floodplains are increasingly vulnerable to climate change. As groundwater levels drop, these habitats become further disconnected from river systems, intensifying the effects of warming on forest ecosystems. A significant rise in temperatures, coupled with increased evapotranspiration, is leading to more frequent and intensive droughts, particularly affecting the topsoil, a critical water source for mature trees with shallow root systems (Szatniewska et al., 2022). These drying trends also adversely impact forest regeneration (Čater and Batič, 2006), potentially undermining long-term species persistence.

Artificial flooding, introduced at the Pohansko site since 2011, provided further evidence for management adaptation. Oak trees responded positively, with increased radial growth and greater hydraulic stability. However, the same intervention showed limited benefits for ash (Bártová et al., 2021). These findings confirm that revitalization measures must be designed with species-specific rooting depths and water uptake capacities in mind.

To sustain and enhance the resilience of floodplain forests under changing climatic conditions, we recommend:

- Artificial flooding through periodic reservoir releases, carefully timed to mimic natural seasonal inundation patterns (Bártová et al., 2021; Klimo and Hager, 2001);

- Deactivation of artificial drainage systems that accelerate groundwater depletion (Klimo and Hager, 2001);
- Establishment of water retention reservoirs that provide a slow and regulated flow of water, which can replicate natural hydrological cycles, maintain groundwater levels, and promote the long-term stability of floodplain forest ecosystems.

At the same time, shifts in species composition are becoming increasingly evident. Climate change and the decline in winter floods have created conditions that favor flood-intolerant, shade-tolerant species such as hornbeam and field maple (Janik et al., 2011). These species now dominate many stands, outcompeting light-demanding species such as oak and ash, and increasing competition for already-limited water resources (Szatniewska et al., 2022). Without active intervention, this transition could reduce structural diversity and weaken ecological function. Forest managers must consider selective thinning, canopy gap creation, and assisted regeneration to maintain the ecological roles of sensitive, historically dominant species.

Historically, South Moravian floodplain forests were intensively managed, but in the past 50 years, forest policy has shifted toward more nature-oriented practices (Klimo and Hager, 2008). While this shift has ecological merit, our findings suggest that a passive approach may be insufficient under rapidly changing hydroclimatic conditions. Trees adapted to nutrient-rich, water-abundant environments may experience reduced growth responsiveness under water stress, particularly in the absence of targeted silvicultural interventions (Stojanović et al., 2015).

6.2 Future management planning for forest adaptation

The higher incidence of negative pointer years in southern oak populations likely reflects increased climatic and biotic stress in these regions (Fig. S6). Although this pattern may be partly influenced by the larger number of plots sampled in the south, the higher frequency of extreme growth reductions in recent decades suggests that southern oak populations are experiencing more frequent and severe climatic disturbances associated with intensified drought conditions (Basu et al., 2024; Mikac et al., 2018). Additionally, anthropogenic river regulations have disrupted groundwater dynamics, further

exacerbating drought stress in lowland areas critical for oak (Basu et al., 2024; Mikac et al., 2018).

Biotic stressors such as insect defoliators and bark beetles likely amplify the frequency and severity of extreme growth reductions observed in southern Oak populations (Haavik et al., 2015; Khasanov and Sandlersky, 2018). These stressors are more common in the southern population of the species' range, where warmer temperatures favour faster insect development and more frequent outbreaks, particularly in recent decades (Milanovic et al., 2023). At the same time, northern populations could also experience increased insect herbivory pressure under warming conditions, since rising winter minimum temperatures may reduce overwintering insect mortality that previously limited outbreaks (Tobin and Raffa, 2022). Southern populations, having evolved under greater biotic pressure, may have developed stronger anti-herbivore defence mechanisms (Moreira et al., 2018), providing an adaptive advantage in a changing climate.

In general, our results suggest that a warmer and drier future climate will lead to significant shifts in Oak populations, particularly along the southern edge of its distribution, where it may eventually be replaced by more drought-resistant oak species (Illés and Móricz, 2022). However, these southern populations may represent valuable sources of drought-adapted genotypes for APM strategies, potentially facilitating faster northward migration than would naturally occur under ongoing climate change (Aitken and Bemmels, 2016; D'Orangeville et al., 2025). Another important risk to consider when planning APM is susceptibility to late frosts (Montwé et al., 2018), to which broadleaved species, particularly ring-porous oaks, are highly sensitive (Vitasse et al., 2019). As mentioned above, both earlywood vessels and leaves are formed using NSC accumulated during the previous growing season (Michelot et al., 2012). Frost damage to newly flushed leaves would require the production of a second flush, leading to substantial carbon costs. This could further weaken the trees, diminishing their ability to withstand subsequent summer droughts or pest outbreaks (Haavik et al., 2015). Some individuals of the Oak subspecies, such as Slavonian oak (*Quercus robur* subsp. *slavonica*), growing near the Sava river in Serbia and Croatia exhibit late flushing but also exceptional wood quality (Burger and Gailing, 2022). Incorporating genetic material from these populations into APM programs

could enhance late-frost resilience while minimizing hybridization and genetic admixture with local oaks (Burger and Gailing, 2022).

On the other hand, high-elevation populations exhibited the highest climate sensitivity and resistance and aligned most closely with the theoretical full-resilience model (Figs. 21, 24, 25). Higher resistance has been recognized as an important trait for reducing drought-related mortality in angiosperm species (DeSoto et al., 2020). Evidently, populations that effectively reduce the initial impact of disturbances (i.e., have higher resistance) inherently demand less recovery effort (Schwarz et al., 2020 and the references therein). Overall, these populations were distinctly different in terms of resistance and recovery (Fig. 24), while southern and northern populations revealed more gradual changes along the gradient. These high elevation populations, situated in the inner Alpine region, were separated from the northern and southern regions by the Carpathians and may differ genetically and/or originate from distinct glacial refugia (Csai et al., 2002). Although our analyses focused on oak, sessile oak is also commonly present in these areas, either forming pure stands or occurring in mixed stands with it, particularly in the sites which are not influenced by groundwater (Wallnöfer and Hotter, 2008). The fact that these two sympatric oak species readily hybridise (Bodénès et al., 1997; Jensen et al., 2009) provides additional support for the observed distinctiveness of these inner Alpine oak populations.

7. Conclusion

This PhD research investigated the influence of prolonged alterations in groundwater levels and climatic variability on the growth, vitality, and hydraulic performance of two predominant Central European floodplain tree species, oak and ash. The work integrates dendroecological, anatomical, and geographic investigations to offer empirical insights into the intricate relationship between hydrological dynamics and tree health, thereby informing future adaptive forest management practices.

The study validated hypothesis H1 by confirming that groundwater availability is a key factor influencing growth sensitivity in both species. Growth responses were tightly correlated with groundwater changes, particularly during dry years when surface precipitation was inadequate. Oak, due to its deeper roots system, demonstrated significant response to subsurface moisture availability, indicating a strategic dependence on deeper soil water reserves. Ash had a stronger correlation with precipitation and surface soil moisture, indicative of its shallower root system and increased susceptibility to drought conditions. These species-specific variations illustrate that groundwater functions not only as a background factor but as a crucial ecological driver, affecting growth patterns and resilience throughout both seasonal and interannual timeframes.

The hypothesis H2, which assumed seasonal variations in tree growth sensitivity to water supply, was also corroborated. development had the highest sensitivity to precipitation in spring, along with earlywood formation, whereas groundwater was more crucial in supporting development during summer and autumn. This seasonal change in water reliance reflects the natural pattern of river runoff and emphasises the ecological need of maintaining both surface and underground water dynamics to ensure sustained forest productivity during the growing season. These findings have obvious implications for the timing of water management interventions, especially artificial flooding, which could be

the most advantageous when synchronised with species-specific phenological requirements.

Evidence for hypothesis H3 further emphasises the enhanced drought tolerance of oak relative to ash. During significant drought years, oak continuously surpassed ash in a number of parameters, such as resistance, recovery, and overall resilience. These divergent tactics demonstrate the interaction of anatomical and physiological features in shaping drought responses, with oak properties promoting long-term survival in drier circumstances.

Consistent with hypothesis H4, ash exhibited greater variability in hydraulic traits than oak, reflecting its more plastic and environmentally responsive nature. This was evident in the dendro-anatomical analyses where trait correlations in ash varied markedly across sites and years, in contrast to oak's more consistent hydraulic profile. Such plasticity may provide ash with flexibility under fluctuating conditions, but it also suggests a higher degree of environmental sensitivity and physiological instability. While plasticity is often considered adaptive, it may potentially compromising resilience under sustained stress.

Hypothesis H5, however, lacked empirical support. Contrary to expectations that ash would benefit more from artificial flooding, the data demonstrated a species-specific reversal: artificially induced spring flooding greatly enhanced oak growth and earlywood hydraulic conductivity, while ash exhibited no discernible response. This discovery is significant, as it alters our comprehension of hydrological treatments. It indicates that oak, rather than ash, is more likely to benefit from regulated water inputs, especially at the onset of the growing season when water availability coincides with vessel construction. These findings underscore the necessity for focused restoration initiatives, rather than presuming uniform advantages all coexisting species.

Likewise, hypothesis H6 lacked support, as overall resilience did not differ significantly among populations. The analysis nevertheless showed that oak populations at higher elevations, which are generally cooler and more humid, displayed the greatest drought resistance, even though southern populations were expected to be more resilient. Southern populations still exhibited adaptive mechanisms that may contribute to resilience under

stress. In contrast, northern populations showed the highest capacity for post-drought recovery, likely linked to local adaptation and a broader range of water-use strategies. These patterns emphasise the role of local environmental history, site elevation, and genetic differentiation in shaping population-level responses.

These observations highlight the significance of site and species-specific, water-sensitive management options that correspond with species' hydraulic characteristics and climate tolerances. Implementing strategic artificial flooding alongside facilitated population migration could establish an effective framework for maintaining floodplain forest resilience amid evolving environmental conditions.

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Appendix A



Figure A1. Sampling site in Lednice in vegetation period of 2022. (Photograph: S. Basu)



Figure A2. Sampling site in Lanzhot in vegetation period of 2023. (Photograph: M. Stojanovic)



Figure A3. Waterlogged site in Soutok in spring of 2023. (Photograph: S. Basu)



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Appendix B

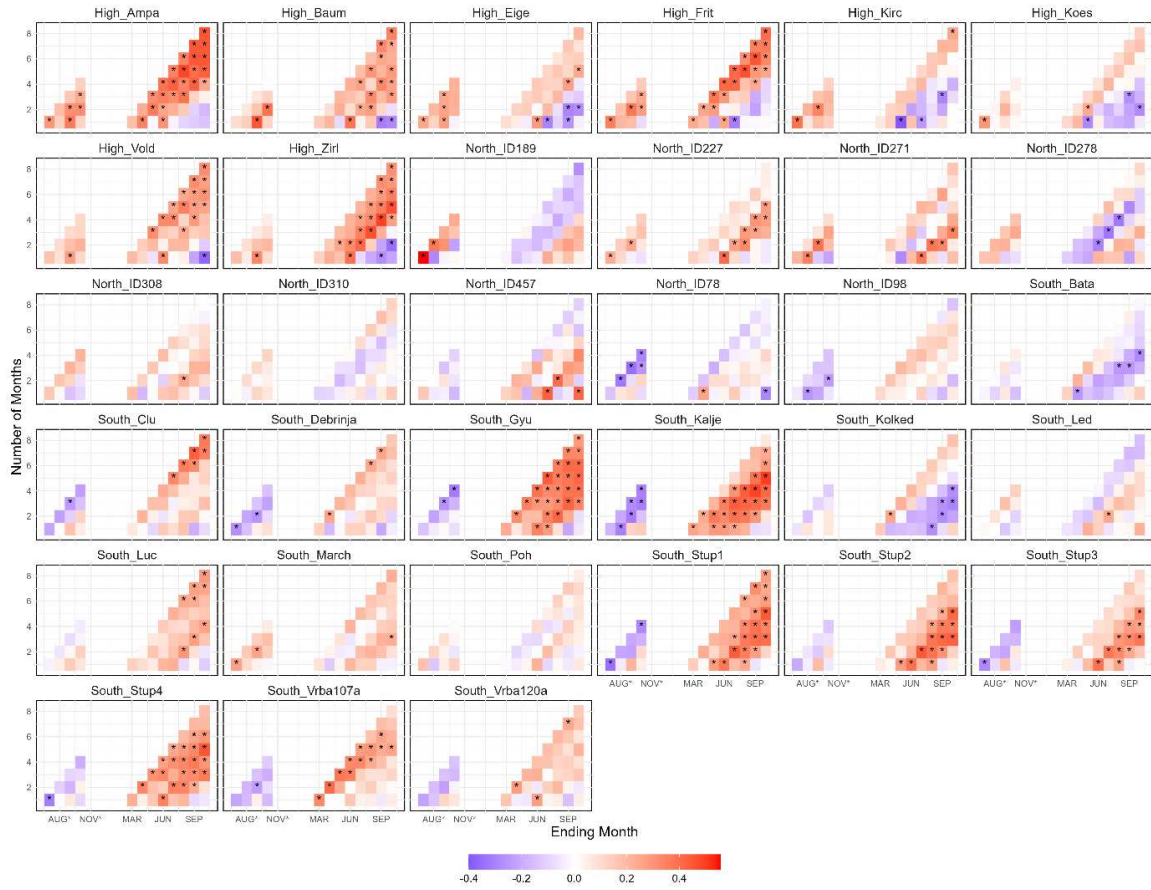


Figure B1. Heatmaps of Pearson correlation coefficients between tree-ring width indices and the standardized precipitation-evapotranspiration index (SPEI) for the 33 study sites. The left side of the panels shows the influence of the previous year's months (July to October) on current growth, while the right side shows the influence of the current year's months (March to October). Analysis considered a 1–4-month integration timescale for the influence of the previous season, while a 1–8-month integration timescale was used to assess the current year's influence on the current year's growth. Red and blue colors represent positive and negative values, respectively, with asterisks marking significant results ($p < 0.05$). Darker shades indicate stronger significance.

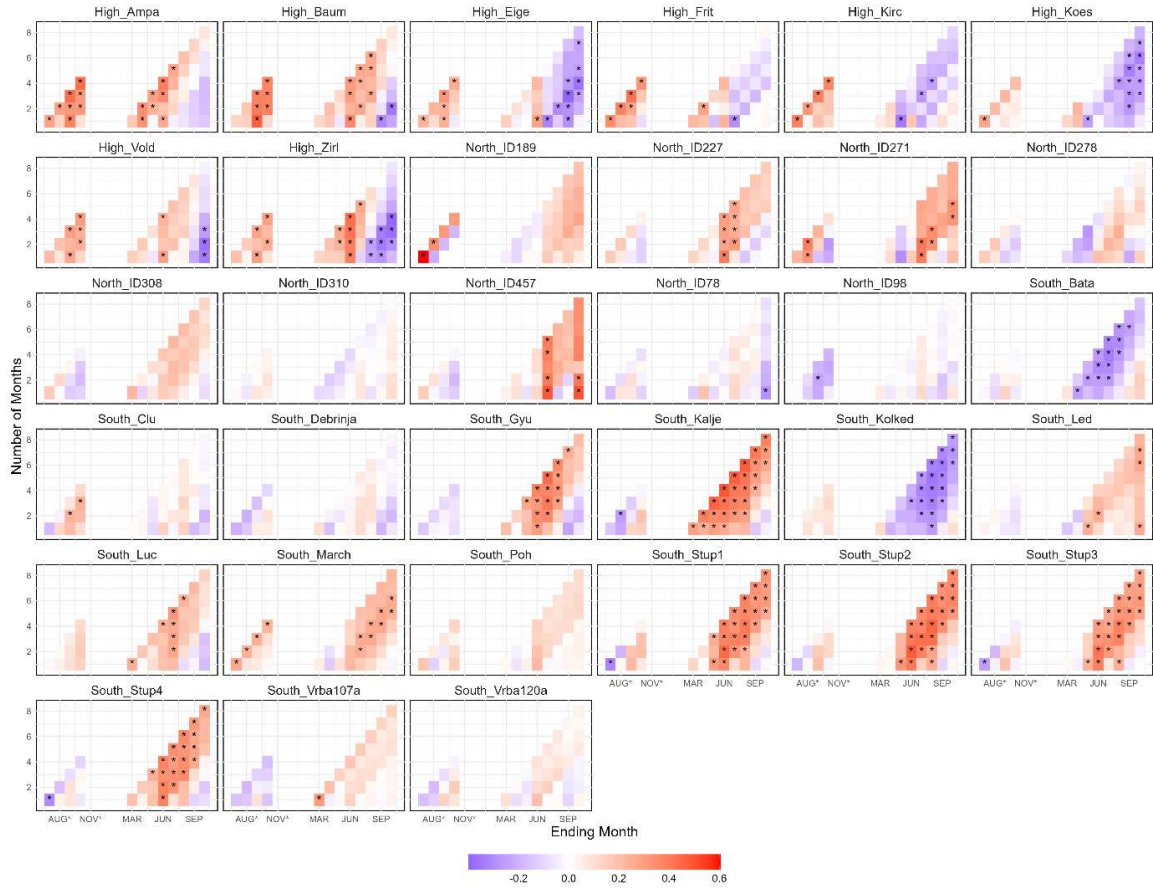


Figure B2. Heatmaps of Pearson correlation coefficients between tree-ring width indices and precipitation (P) for the 33 study sites. For a detailed description of the figure, please refer to Fig. B1.

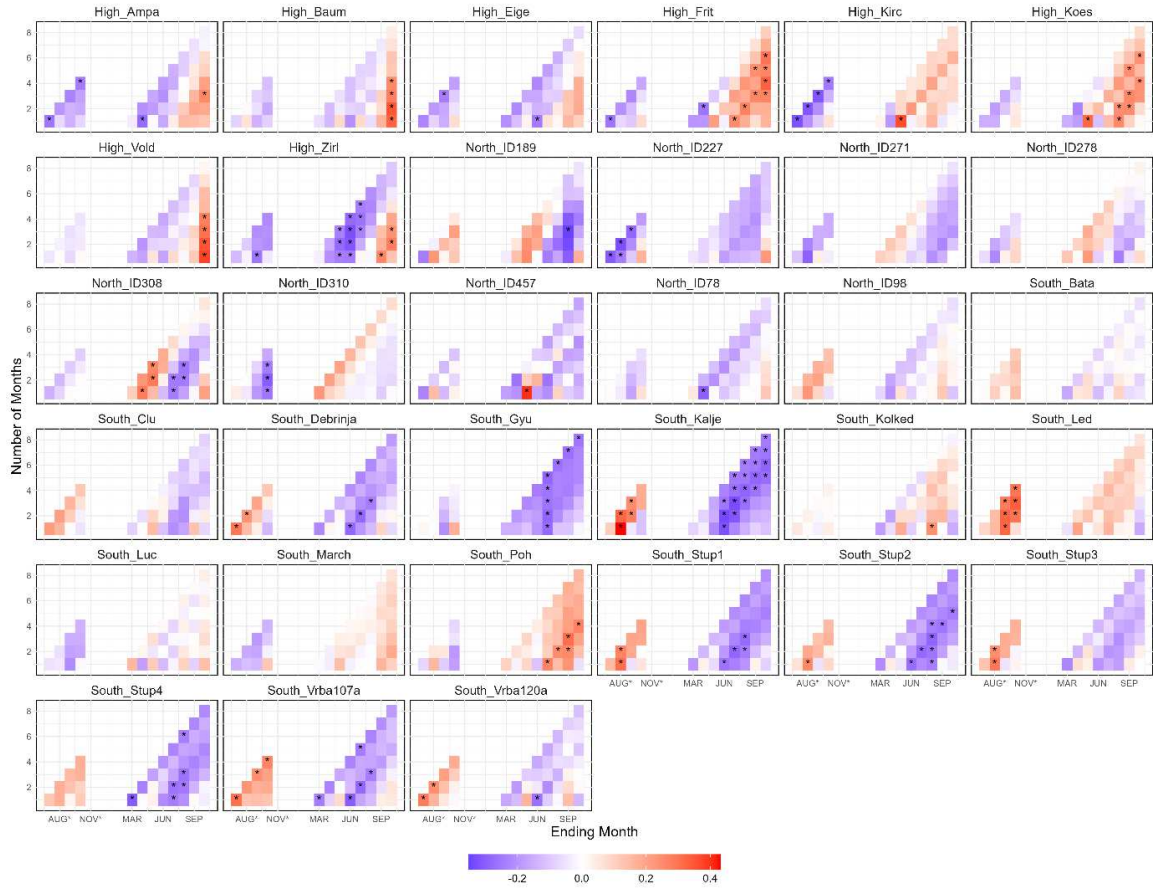


Figure B3. Heatmaps of Pearson correlation coefficients between tree-ring width indices and maximum temperature ($T_{\max}$) for the 33 study sites. For a detailed description of the figure, please refer to Fig. B1.

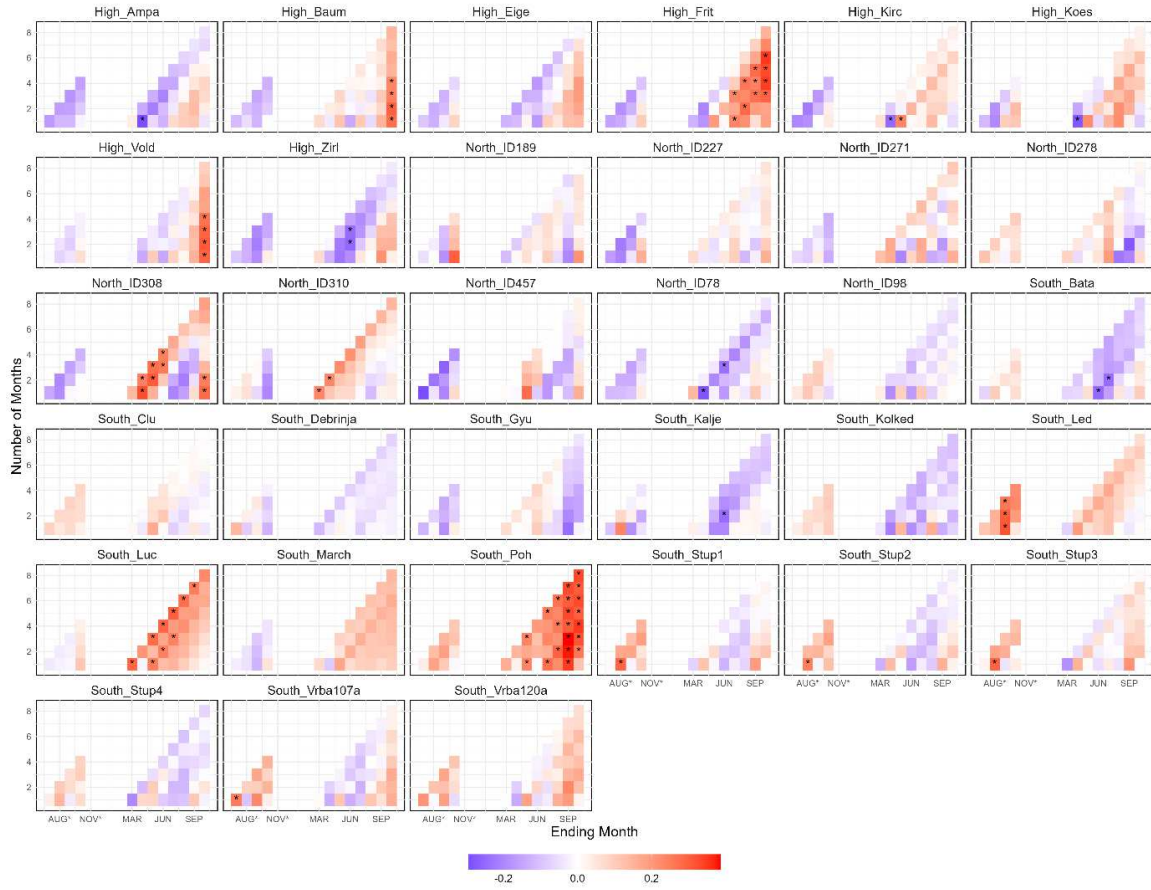


Figure B4. Heatmaps of Pearson correlation coefficients between tree-ring width indices and minimum temperature ($T_{\min}$) for the 33 study sites. For a detailed description of the figure, please refer to Fig. B1.

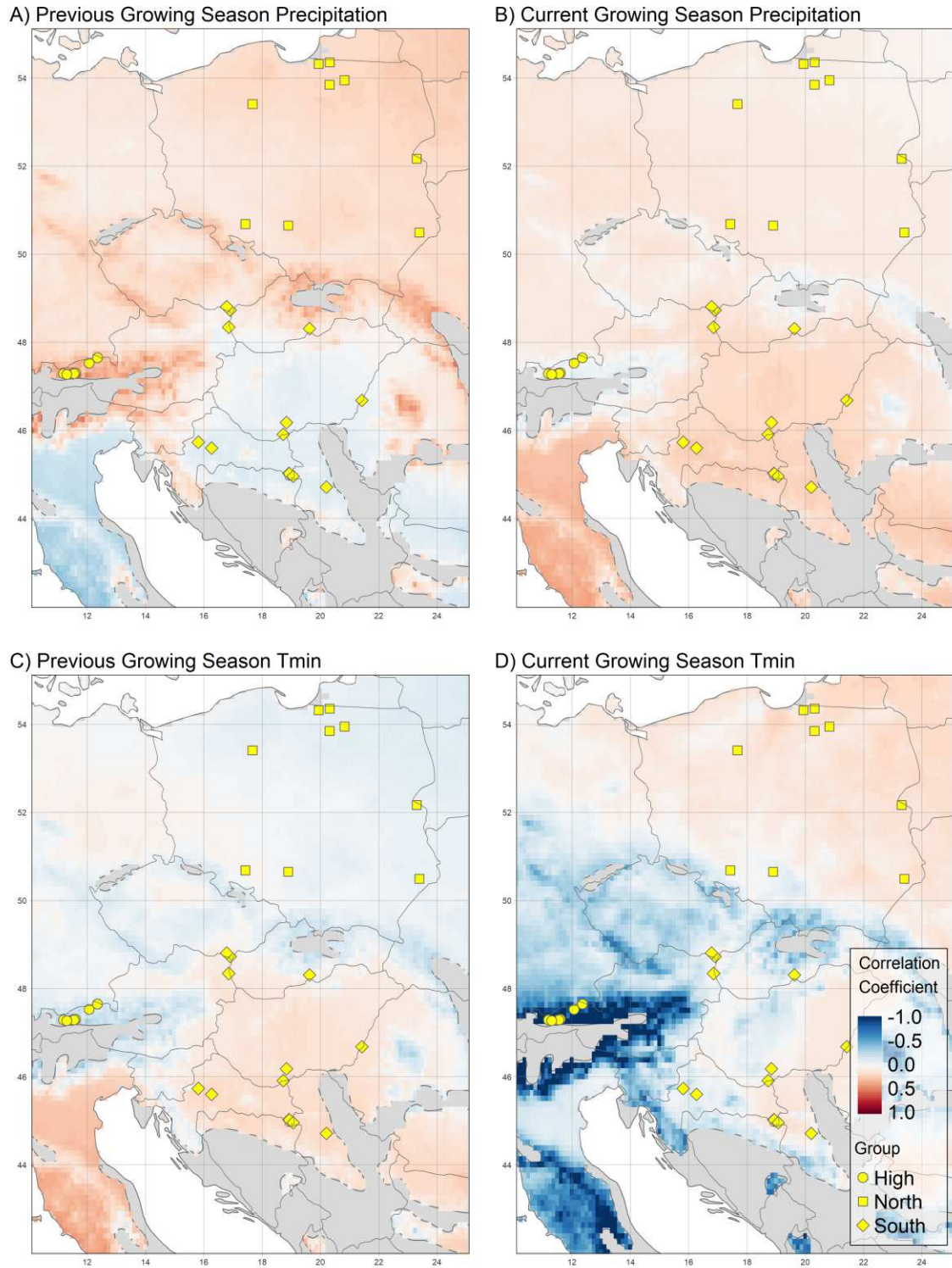


Figure B5. Spatial patterns of *Quercus robur* climate sensitivity. The upper panels show the influence of previous (A) and current (B) precipitation, while the lower panels show the influence of previous (C) and current (D) minimum temperature ($T_{\min}$). Symbol shapes indicate cluster grouping: South – diamond, North – square, and High – circle.

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Climate	Season	Term	Estimate	p -value	Adjusted R^2	BIC
Precipitation	Current Growing Season	(Intercept)	-0.26	0.23	0.06	27.07
		MAT	0.04	0.08		
	Previous Growing Season	(Intercept)	0.69	0.00	0.37	-0.27
		MAT	-0.07	0.00		
SPEI	Current Growing Season	(Intercept)	-0.15	0.46	0.06	22.36
		MAT	0.04	0.08		
	Previous Growing Season	(Intercept)	0.89	0.00	0.62	-11.21
		MAT	-0.09	0.00		
$T_{\max}$	Current Growing Season	(Intercept)	0.47	0.009	0.22	10.79
		MAT	-0.06	0.003		
	Previous Growing Season	(Intercept)	2.74	0.00	0.66	-20.17
		Elevation	0	0.00		
		Latitude	-0.04	0.00		
		Longitude	-0.03	0.03		
$T_{\min}$	Current Growing Season	(Intercept)	5.64	0.006	0.22	6.57
		MAT	-0.18	0.002		
		Elevation	0	0.002		
		Latitude	-0.07	0.01		
	Previous Growing Season	(Intercept)	-0.49	0.00	0.47	-24.93
		MAT	0.06	0.00		

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Table 5: Summary of Generalized Additive Model (GAM) results for the effects of environmental and temporal variables on mean lumen area (MLA) and hydraulic diameter (Dh) for ash and oak.

Abbreviations

Abbreviation	Full Term
AIC	Akaike Information Criterion
ANOVA	Analysis of Variance
APM	Assisted Population Migration
Ar1	First-order Autocorrelation
BIC	Bayesian Information Criterion
CD	Conduit Diameter
DBH	Diameter at Breast Height
Dh	Hydraulic Diameter
Edf	Estimated Degrees of Freedom
EPS	Expressed Population Signal
EWV	Earlywood Width
GAMM	Generalized Additive Mixed Model
GLK	Gleichläufigkeit
Kh	Hydraulic Conductivity
LWW	Latewood Width
MAP	Mean Annual Precipitation
MAT	Mean Annual Temperature
MLA	Mean Lumen Area
MS	Mean Sensitivity

NSE	Nash–Sutcliffe Efficiency
NSC	Non-Structural Carbohydrates
PCA	Principal Component Analysis
PC1	First Principal Component
PC2	Second Principal Component
PCGA	Principal Component Gradient Analysis
PET	Potential Evapotranspiration
Rbar	Mean Interseries Correlation
REML	Restricted Maximum Likelihood
r rmax	Maximum Correlation Coefficient
SPEI	Standardized Precipitation Evapotranspiration Index
SPEI3	3-Month Standardized Precipitation Evapotranspiration Index
SPEI12	12-Month Standardized Precipitation Evapotranspiration Index
SNR	Signal-to-Noise Ratio
Tmax	Maximum Temperature
Tmean	Mean Temperature
Tmin	Minimum Temperature
Trwi	Tree-Ring Width index
TRW	Tree-Ring Width
VPD	Vapor Pressure Deficit